Group flights in free flight

Delayed contacts, configurations and episodes:
a method to label every instant of every flight

Matteo Di Sante

Method note, working version, 23 October 2026
English version, with figures and animations, of the Italian note of 14 October 2026
In red: corrections of 22 October 2026, from a check of the code against this note.

Abstract

We describe a method that decides, for every flight and every instant, whether a paraglider pilot flies in a group or alone and, in the first case, with which other pilots. Two pilots are in contact when one of them is where the other was, or will be, a few minutes earlier or later. This is the proximity with a local delay of Li, Wu and Crofoot [15], adapted to thermal flight in three ways: a vertical tolerance separates pilots who circle in the same thermal at distant heights; headings are compared only in glide; the maximum delay is bounded by the visual range and measured on the data. At each instant the groups are the connected components of the contact graph. As in the grouping structure of Buchin et al. [6], a configuration is a fixed set of flights over a maximal interval, and an episode is a chain of configurations that share members. Contacts are found in two stages, a coarse filter that provably loses none and an exact scan; their minimum duration is calibrated on a null model that shifts tracks in time. On the FFVL archive most short contacts are chance: the maximum delay is measured at 2 min, and a contact must last 38 min for at most one in ten to be chance. With these values 21.7% of the paraglider flights and 6.9% of the hang-glider flights fly in a group at some point, in 17 169 episodes. The horizontal and vertical tolerances show no plateau on the data and are declared as definitions. The output is a few tables that label every instant of every flight, so that group flight can be compared with solo flight.

Contents

The method at a glance **3**

1 Motivation and requirements **4**

2 Relation to the literature **5**
2.1 Trajectory similarity with a temporal tolerance 5
2.2 Following with a delay 5
2.3 Group structure in time 6
2.4 Interaction indices and null models 6
2.5 Collective soaring 7
2.6 What we take and what is new 7

3 Data and notation **8**

4 The pair contact **9**
4.1 Local phase . . . 9
4.2 Correspondences and contact . . . 10
4.3 The role of each term . . . 12
4.4 Choice of the maximum delay . . . 13
4.5 How far behind: the delay of a contact . . . 13

5 From pairs to groups **15**
5.1 Pair intervals . . . 15
5.2 The problem of non-transitivity . . . 16
5.3 Configurations . . . 17
5.4 Episodes . . . 18
5.5 Dependency blocks . . . 19
5.6 An example day . . . 19

6 Algorithm **19**
6.1 How a run is organised . . . 21
6.2 Phase 0: preparation per flight . . . 21
6.3 Phase 1: candidate pairs . . . 22
6.4 Phase 2: exact correspondences . . . 25
6.5 Phase 3: pair intervals . . . 26
6.6 Phase 4: configurations and episodes . . . 27
6.7 Costs and space . . . 27

7 Calibration: what chance means **28**
7.1 The null model . . . 28
7.2 The false discovery rate and τ^* . . . 29
7.3 How much the shift matters . . . 31

8 The parameters on the data **31**
8.1 The maximum delay δ . . . 31
8.2 The tolerances against chance . . . 34
8.3 One parameter at a time . . . 34
8.4 The false discovery rate q . . . 35
8.5 Fixed settings . . . 36

9 The groups on the archive **36**

10 Outputs **37**

11 Use for the comparison between group and solo flight **38**

12 Choices to declare and limits **39**
12.1 Other checks . . . 40

A Why $m_{ij}(t) = 1$ does not give $m_{ji}(t) = 1$ **40**
A.1 A worked example . . . 40
A.2 The same example in the plane (t, t') . . . 40
A.3 How far apart they can be . . . 41
A.4 Why the contact is the union . . . 42

The method at a glance

Figure 1 follows the method from two tracks to the units of the statistical analysis, and Table 1 says what each object is in plain words, with the section that defines it. Figures 7, 12 and 18, marked ▶ , are animated: in Adobe Acrobat Reader, or another viewer that runs the JavaScript of PDF files, the buttons below them play the animation; other viewers show the figure as printed. The same animations are GIF files in the folder `animations/`.

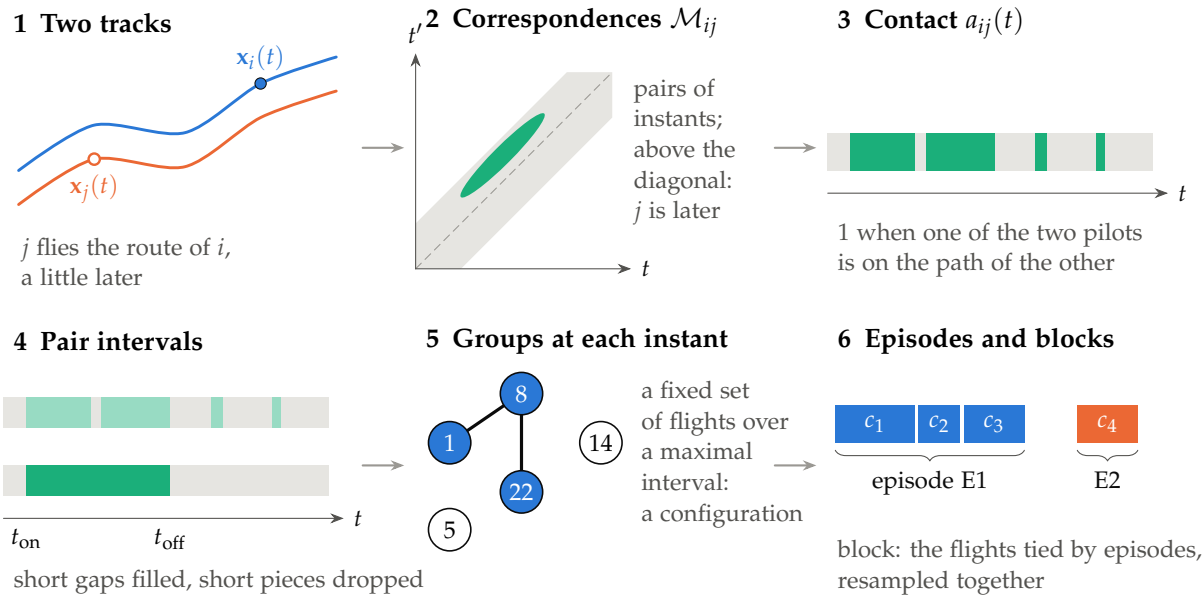


Figure 1: The method in six steps. Panels 1–3: the pair contact (Section 4). Panels 4–6: from pairs to groups (Section 5). Flights 1, 8, 22, 14 and 5 are the example day of Figure 18.

Object	In words	Symbol	Section
Departure	the first fix farther than R_{dep} from the takeoff	k_i^{dep}	4.1
Correspondence	two instants of the same clock, t for i and t' for j, at most δ apart, at which the two pilots pass the same place within the tolerances	$(t, t') \in \mathcal{M}_{ij}$	4.2
Contact	at instant t, one of the two pilots is on the path of the other	$a_{ij}(t) = 1$	4.2
Pair interval	a stretch of contact, with short gaps filled and short pieces dropped	$[t_{\text{on}}, t_{\text{off}})$	5.1
Group at instant t	the pilots joined to each other by chains of contacts at t	component of $G(t)$	5.3
Configuration	a fixed set of flights that stays one group over a maximal interval	c	5.3
Membership	the label of flight i at instant t: local phase, solo, or its configuration	$\gamma_i(t)$	5.3
Episode	a chain of configurations, each starting when the previous one ends and sharing a member with it		5.4
Dependency block	the flights tied together by episodes; the bootstrap resamples them together		5.5

Table 1: A minimal dictionary of the objects of the method.

1 Motivation and requirements

The statistical properties of free-flight trajectories (time spent in thermals, length of the glides, mean squared displacement) are usually estimated by treating flights as independent samples. The assumption fails when several pilots fly together: they share thermals and routes, and their trajectories are correlated. Group flight may also change the behaviour itself. In soaring birds, social information on the position of thermals changes the gliding strategy [28], and in a migrating flock a few individuals find the thermals while the others follow them [8, 19]. Comparing group flights with solo flights therefore needs a rule that says, for every flight and every instant, whether the pilot is in a group and with whom.

The intuitive notion of *flying together* that the rule must make operational is the following (Figure 2).

(R1) The pilots take off from the same area, not necessarily from the same takeoff.

(R2) Departures may be staggered, but not to the point of losing sight of each other.

(R3) The pilots follow roughly the same route: nearby places and similar motion, not necessarily the same points.

(R4) The composition of the group changes in time: some pilots join, others change route.

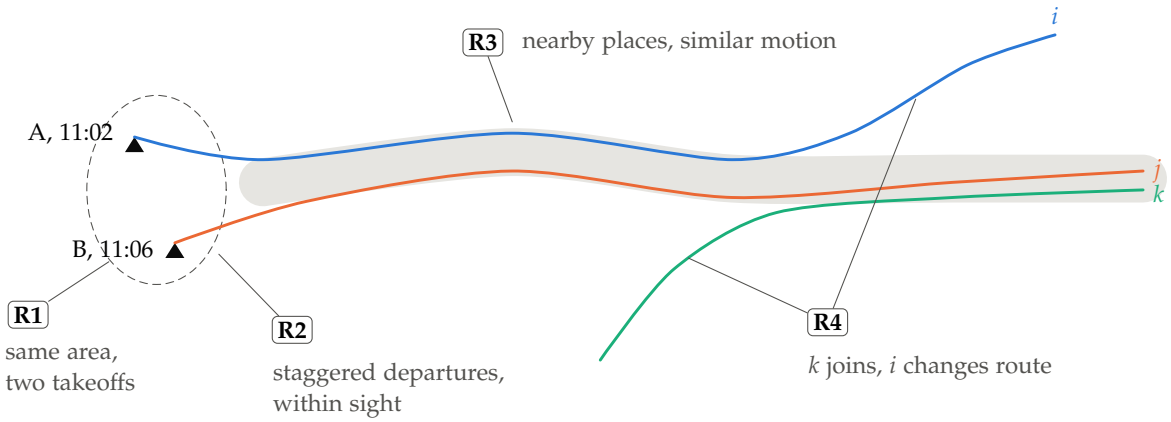


Figure 2: The four requirements on a schematic map. Pilots i and j take off from two places of the same area four minutes apart and follow the same corridor (grey) without flying over the same points; k joins later and i leaves the corridor.

Three immediate approaches do not meet these requirements (Figure 3).

- **Geometric similarity of the tracks** (Hausdorff, Fréchet, unconstrained DTW; survey in [23]). Cross-country routes are channelled by the terrain and thermals are often anchored to the ground: two independent pilots who took off from the same site produce similar tracks even hours apart. Unconstrained DTW, moreover, warps the time axis without limit and erases exactly the information that (R2) needs.
- **Strict synchrony** (flock [10], convoy [11], swarm [14]). These patterns require proximity at the same instant; the swarm admits non-consecutive instants but no delay between members. Two pilots on the same route two minutes apart are never close at the same instant, yet they fly together in the sense of (R2)–(R3).

- **Window on the departures.** A rule such as "departures within T minutes" produces groups that depend on the reference pilot: if 1 and 8 depart within the window, and so do 8 and 14, then 1 and 14 need not (Section 5.2).

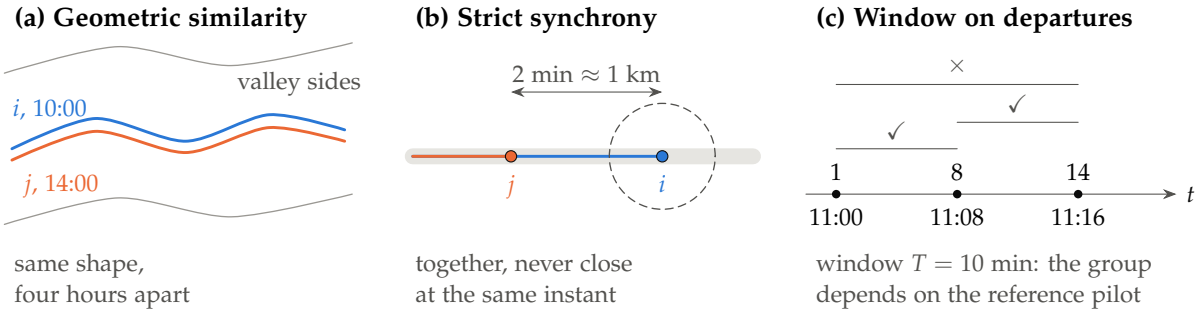


Figure 3: Why three immediate rules fail. (a) Two independent pilots on the same valley produce similar tracks hours apart. (b) Two pilots on the same route two minutes apart never enter each other's proximity disc. (c) With a window of 10 minutes, 1–8 and 8–14 pass and 1–14 fails.

2 Relation to the literature

This section says where each ingredient of the method comes from and what, as far as a targeted but not exhaustive search shows, has not been done yet.

2.1 Trajectory similarity with a temporal tolerance

The LCSS of Vlachos, Kollios and Gunopulos [27] matches two points when they are closer than ε in every coordinate and their *indices* in the two sequences differ by at most δ. The similarity S_1 is the length of the common subsequence divided by the length of the shorter track, and the variant S_2 maximises S_1 over all spatial translations, to recognise parallel movements. The EDR of Chen, Özsu and Oria [7] and DTW with a Sakoe–Chiba band [21] bound the shift in the same way. These measures return a single number per pair of tracks: they say how similar two tracks are, and not *when* two pilots are together. A bound on indices counted from the first point of each track also mixes the shift between the takeoffs with the delay in flight, and S_2 matches precisely the parallel routes that we want to exclude.

The measures of joint movement that do say when are built on simultaneous fixes. The proximity index Prox is the share of simultaneous fixes closer than a threshold [12, §3.1]: with $\delta = 0$ and without the conditions on height and heading, it is the share of the flight of i in contact with j in the sense of Section 4.2.

2.2 Following with a delay

The closest precedent of our pair contact is the following of Li, Wu and Crofoot [15], defined on tracks sampled at the same instants (Figure 4). At instant i the local lag $\Delta(i)$ is the shift t, with $|t| \leq l_{\max}$, that minimises the distance between the position r_i of R and the position s_{i-t} of S. The instant is *valid* if this minimum distance is below $d_{\max}$, and R follows S at a valid instant if $\Delta(i) > 0$ (§IV-A). The lag is therefore local and free to vary, and its sign separates the follower from the leader. Two individuals that merely move together lead and follow equally often, so the following intervals are the stretches with more following instants than the half expected without following: the maximal segments of $f(i) - 1/2$, with $f(i) = 1$ when R follows and 0

otherwise, found in linear time [20] (§IV-B) and checked with the roles reversed (§IV-C). The threshold $d_{\max}$ only discards the instants without interaction, and does not move the minimiser.

Our contact keeps the window of lags and the distance threshold, and leaves the sign out: the contact of i with j at instant t is the validity of t, with tolerances on height and heading added (Section 4.2). We want the pairs that fly together, whether or not one of them leads; who follows whom comes back afterwards, through the median lag of each contact (Section 4.5).

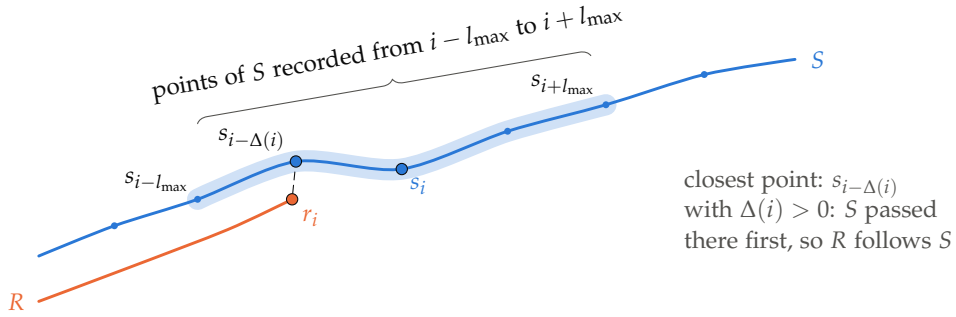


Figure 4: Following with a local lag [15]. At instant i, the track of S is searched in the time window $[i - l_{\max}, i + l_{\max}]$ (shaded) for the point closest to the position r_i of R. The best lag $\Delta(i)$ is positive: R is now where S was $\Delta(i)$ earlier.

Nagy et al. [18] measure the delay between two pigeons as the shift that maximises the correlation of their headings, and build leader–follower hierarchies from its sign. In a thermal the heading turns through 2π every 20–30 s, so this correlation, like the distance of Section 4.3, repeats with the period of the turn. Konzack et al. [13] display such local delays between simultaneously recorded tracks in a "delay space", comparing Fréchet, DTW and edit-distance matchings. FLICA and its package mFLICA [2, 3, 4] infer a following network in sliding time windows, with DTW restricted to a Sakoe–Chiba band, and call the faction of a leader the individuals that reach it through a directed path: groups built from delayed following, but directed, with leaders, and on windows. The following of Andersson et al. [5] is instantaneous: an individual follows another that is in a sector ahead of it with a similar heading.

2.3 Group structure in time

Buchin et al. [6] call two entities directly connected at time t if their ε-discs overlap, and ε-connected if a chain of direct connections joins them at that instant; the maximal ε-connected sets, the components, partition the entities at every instant. The Reeb graph records how the components evolve: its vertices are the events (start, end, merge, split), and each edge is a fixed set of entities that forms one component over an interval of time (Figure 5). On it they define the maximal groups, sets of at least m entities that stay inside one component for at least a duration δ; maximal groups may overlap, and an entity can belong to several at once. Two more ideas of theirs reappear below: a robust structure ignores interruptions shorter than α, by calling two entities connected at t when they were directly connected at some instant within $\alpha/2$ of t (§4); and the grouping structure can split the trajectories into independent data sets (§1). The proximity is always simultaneous.

2.4 Interaction indices and null models

Joo et al. [12] sort the metrics of joint movement between pairs by what they capture, proximity or coordination. Long et al. [16] compare eight of them on simulated tracks and on deer: the indices that rest on a statistical test report interactions that are not there, more often the finer

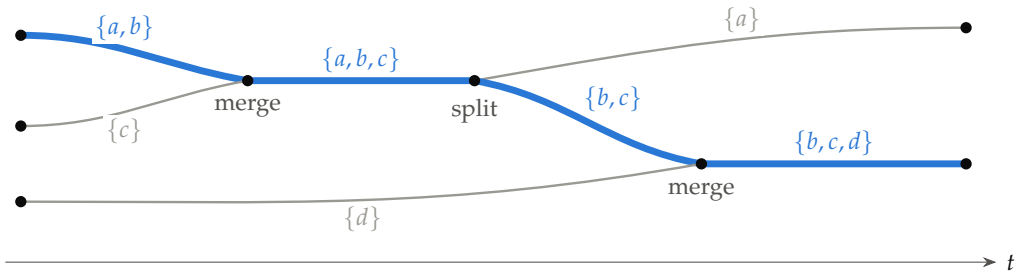


Figure 5: The grouping structure of Buchin et al. [6] as a Reeb graph. Each edge is a fixed set of entities that forms one connected component over an interval; vertices are starts, ends, merges and splits. The thick edges have at least two entities: in this note they become the configurations (Section 5.3), built on delayed contacts.

the sampling, and the indices that return one number per pair miss the interactions that come and go; the wildlifeDI package [17] extends the analysis to contacts in high-frequency data. Spiegel et al. [22] build a null model by shuffling whole days of each individual's path among its days: each individual keeps its use of space, and the synchrony between individuals is lost.

2.5 Collective soaring

Ákos, Nagy and Vicsek [1] compare the soaring of birds with that of paraglider and hang-glider pilots on GPS tracks; falcons, storks and paragliders circle with radii of 20–26 m. In migrating white storks a few leaders find and explore the thermals and the others follow them, leaving the thermals earlier and lower [8]; inside a thermal, flock mates synchronise their circling [19]. Vultures informed by other birds of the next updraft glide towards it faster [28]. We found no work that identifies group flights systematically in large free-flight datasets.

2.6 What we take and what is new

The method is not a new general algorithm for grouping trajectories. The pair contact is the proximity with a local lag of Li et al. [15], the group structure is that of Buchin et al. [6], and groups built from delayed relations exist in mFLICA [4], with other definitions (Table 2). What we did not find in the literature is:

(i) the Reeb structure built on contacts with a local delay instead of simultaneous proximity;

(ii) the adaptations to thermal flight: a vertical tolerance that separates pilots circling in the same thermal at distant heights, and headings compared only in glide (Section 4.3);

(iii) the application to a free-flight dataset of this size, with thresholds calibrated on a null model.

The contribution is therefore a physically motivated adaptation and a scalable implementation, more than a new method in itself, and the thesis should present it in these terms.

Essential reading. The two readings to do first are Li et al. [15], Sections II and IV (following, local lag, score), and Buchin et al. [6], Sections 1, 2.1 and 4 of the arXiv version (groups, Reeb graph, robustness). Long et al. [16] and Spiegel et al. [22] serve the null model; Vlachos et al. [27] and Joo et al. [12] place the measures of similarity and proximity.

Component	Precedent	What we take	What we change
Pair contact with a delay	Li et al. [15]; tolerance in [27, 7, 21]	window of lags and distance threshold; local lag, free to vary	height tolerance (separates heights within a thermal); heading only in glide; local phase excluded; no sign in the contact, $a_{ij} = a_{ji}$ (the sign returns in ℓ_{med})
Contact intervals	robustness in [6]; maximal segments in [15]	tolerance to short interruptions	gaps shorter than g filled; minimum duration calibrated on a null model
Groups in time	Buchin et al. [6]	components and their Reeb graph; configurations are its edges with at least two entities	built on delayed contacts; episodes are the connected parts of the Reeb graph without single entities; no maximal groups
Groups from following	mFLICA [4]	–	undirected groups without leaders; exact intervals instead of sliding windows
Null model	Spiegel et al. [22]; SAM [26]	time shift of the tracks; false discovery rate as chance count over observed count	shift within the day, drawn from the real delays between takeoffs; chance count per unit of exposure
Computation	filter–refine scheme	–	no-loss lemma for delayed contacts; KD-tree with a time coordinate

Table 2: Where the ingredients of the method come from.

3 Data and notation

The flights are those of the xc-dynamics pre-processing. Each flight $i \in V$ is one or more segments of fixes, from its takeoff k_i^0 to its landing k_i^{fin} (step 8 of the pre-processing, take-off and landing); between two segments the track has a gap. A fix has a time, a latitude φ, a longitude λ and the GNSS altitude z, at the interval of its logger: one second for 70% of the paraglider flights, two to ten seconds for most of the others. Time k is in UTC seconds from the start of the day, on the same clock for all flights; IGC times are whole seconds.

Positions follow the frame of the thesis (chapter Trajectory pre-processing, section Local coordinates and vertical datum): each fix is moved to the point of the WGS84 ellipsoid below it, $h = 0$, and converted to Earth-centred Cartesian coordinates, $\mathbf{x}_i(k) \in \mathbb{R}^3$. The thesis then rotates them into the east and north of the first fix of each flight; here they stay Earth-centred, so the coordinates of all flights are the same frame. The distance between two positions is the chord of the ellipsoid, shorter than the distance along it by about $d^3/24R^2$: about a millimetre at $d = 10$ km. The altitude is z as recorded.

The phases of the run reported here are Vilpellet's runs, computed by the old repository (soaring, tag `legacy-final`) on the flights logged every second only, 70% of the paraglider flights. The phase labelling of xc-lab (note *Flight phases with a hidden Markov model*), meant for loggers of 5 s or less, was not used for this run. Every labelled fix of a flight has one of three states, $s_i(k) \in \{\mathsf{climb}, \mathsf{search}, \mathsf{transition}\}$: climbing in a thermal, searching for the next one, gliding between two thermals. In the text *glide* means the state transition. The fixes of all the

other flights and of pieces too short to be labelled stay unclassified.

Symbol	Meaning	Start	Chosen
$\mathbf{x}_i(k) \in \mathbb{R}^3$	position: the point of the ellipsoid below the fix, Earth-centred		
$z_i(k)$	GNSS altitude		
$s_i(k)$	state: climb, search, transition (glide), or unclassified		
$\hat{\mathbf{v}}_i(k)$	horizontal heading, used only in glide		
k_i^0, k_i^{fin}, k_i^{dep}	takeoff, landing, departure (Section 4.1)		
T_i^{dep}	the grid instants from departure to landing at which flight i has a position		
R_{dep}	radius of the local phase around the takeoff	3 km	3 km
Δ	step of the common time grid $\Delta\mathbb{Z}$	5 s	5 s
$h_{\max}$	longest time between two fixes across which a grid position is interpolated	10 s	10 s
δ	maximum delay between two pilots, always a time	5 min	2 min
ε_h, ε_z	horizontal and vertical tolerances	500 m, 300 m	500 m, 300 m
$\theta_{\max}$	maximum angle between the headings in glide	45°	45°
g	longest gap filled in a contact	120 s	120 s
τ^*	minimum duration of a contact	from the null model	38.1 min
v_{xc}	mean cross-country speed, measured on the data	≈ 5 m/s	
$v_{\max}$	top speed between two fixes, set by the pre-processing (`speed.max_kmh`)	100 km/h	100 km/h
ε_{vis}	visual range	1.5 km	
$r_{\min}$, R_s	smallest time shift and radius of the takeoffs in the null model	$> 2\delta$, 10 km	$2\delta + \Delta$, 10 km
q	false discovery rate accepted for the contacts	0.05	0.1

Table 3: Symbols and parameters, with their starting values and the values chosen on the data (Section 8). Beware: in [27] δ is a tolerance on indices, in [12] a distance threshold, in [6] a minimum duration; here it is always a delay.

4 The pair contact

4.1 Local phase

Near the takeoff every pilot is close to every other, ridge soaring or waiting for the thermals, and a contact in this phase says nothing about (R3). The departure is the first fix farther than R_{dep} from the takeoff:

$$k_i^{\text{dep}} = \min\{k : \ \|\mathbf{x}_i(k) - \mathbf{x}_i(k_i^0)\| > R_{\text{dep}}\}. \tag{1}$$

From then on the flight is evaluated on T_i^{dep}, the instants of the common grid, one every Δ seconds, from the departure to the landing at which the flight has a position (Section 6.2). The instants before the departure are not evaluated, and a flight that never leaves the disc is *local*. Both get the state L and do not count as solo flight (Figure 6).

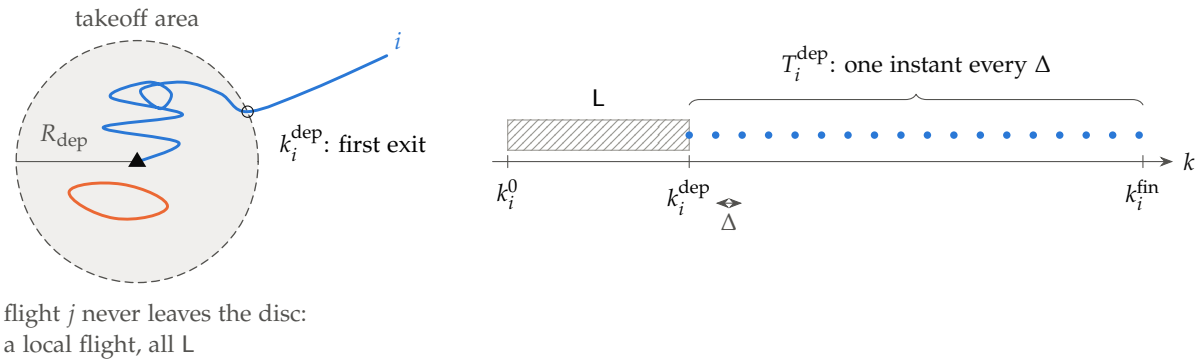


Figure 6: Local phase. Left: near the takeoff the pilot ridge soars and circles inside the disc of radius R_{dep}; the departure k_i^{dep} is the first fix outside it. A flight that never leaves the disc is local. Right: the same flight in time; before the departure the state is L, after it the flight is evaluated on the grid instants T_i^{dep}.

4.2 Correspondences and contact

Definition 4.1 (correspondence)**.** For two flights $i \neq j$, the set of correspondences is

$$\mathcal{M}_{ij} = \Big\{ (t,t') \in T_i^{\text{dep}} \times T_j^{\text{dep}} : |t'-t| \le \delta,\ \ \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| < \varepsilon_h, \\ |z_i(t) - z_j(t')| < \varepsilon_z,\ \ D_{ij}(t,t') = 1 \Big\}, \tag{2}$$

where the heading condition is active only when both pilots glide:

$$D_{ij}(t,t') = \begin{cases} \mathbf{1}\big[\hat{\mathbf{v}}_i(t) \cdot \hat{\mathbf{v}}_j(t') > \cos\theta_{\max}\big] & \text{if } s_i(t) = s_j(t') = \text{transition}, \\ 1 & \text{otherwise.} \end{cases} \tag{3}$$

> **In words.** $\mathcal{M}_{ij}$ is a set of *pairs of instants*. The pair (t,t') belongs to it when j, at its instant t', is where i is at its instant t: within ε_h horizontally and ε_z vertically, with the two instants at most δ apart, and heading the same way if both glide. In the (t,t') plane of Figure 7 it is a region inside the band $|t'-t| \le \delta$.

Definition 4.2 (contact)**.**

$$m_{ij}(t) = \mathbf{1}\big[\exists\, t' :\ (t,t') \in \mathcal{M}_{ij}\big], \quad m_{ji}(t') = \mathbf{1}\big[\exists\, t :\ (t,t') \in \mathcal{M}_{ij}\big], \quad a_{ij}(t) = m_{ij}(t) \vee m_{ji}(t). \tag{4}$$

The conditions in (2) are symmetric, so $\mathcal{M}_{ji}$ is $\mathcal{M}_{ij}$ with the coordinates swapped and $a_{ij} = a_{ji}$. The two halves m_{ij} and m_{ji} differ in general; Appendix A shows why, on a worked example.

> **In words.** $m_{ij}(t) = 1$ when, at instant t, pilot i is at a place that j visits within δ, before or after; $m_{ji}(t) = 1$ when, at instant t, pilot j is at a place that i visits within δ. Their union $a_{ij}(t)$ is the symmetric indicator of contact between i and j at time t: at least one of the two is on the path of the other, within the delay δ.

For the distance alone, $m_{ij}(t) = 1$ is the validity of instant t in the sense of Li et al. [15, §IV-A] (Section 2.2): the distance minimised over the delays is below ε_h exactly when some delay brings it below. The vertical tolerance, the heading condition tied to the flight phase and the symmetry of a_{ij} are ours; the sign of the delay remains available to tell who follows whom (Section 4.5).

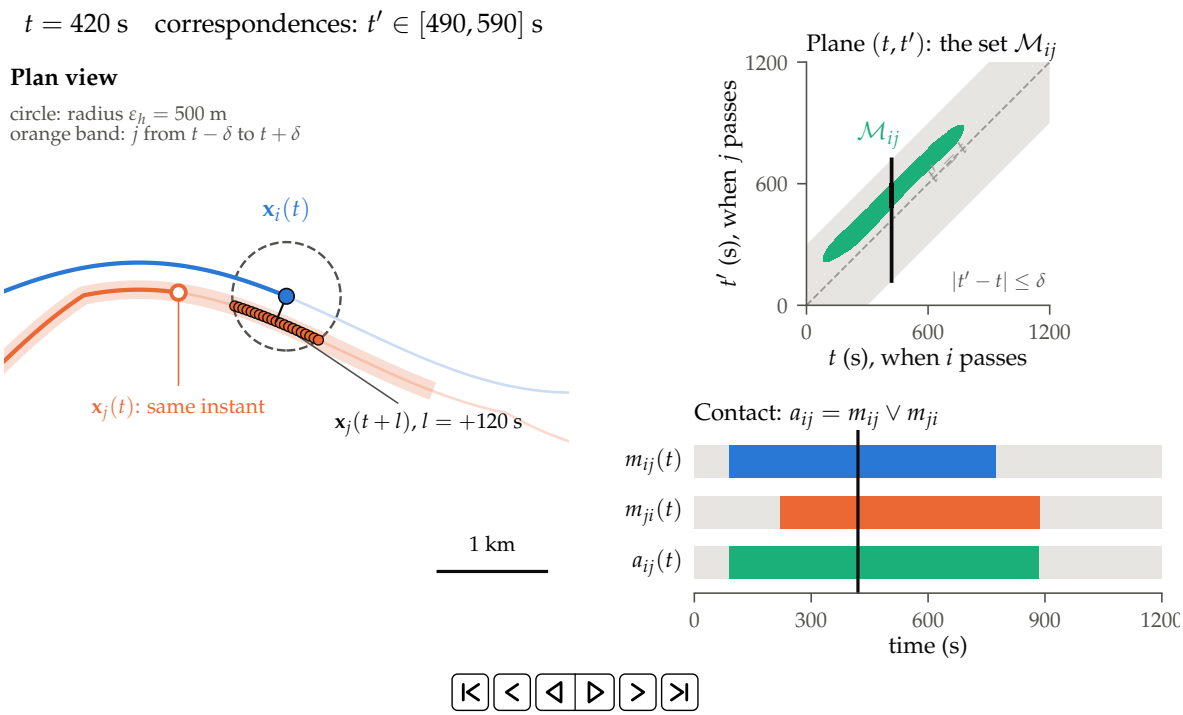


Figure 7: ▶ Correspondences and contact for a synthetic pair; the animation runs the instant t from 0 to 1200 s. Left: plan view around i at $t = 420$ s. Pilot j flies the route of i 120 s later and 250 m to the side; the circle has radius ε_h; the orange band is the part of the track of j flown between $t - \delta$ and $t + \delta$; the dark dots are the instants t' with $(t, t') \in \mathcal{M}_{ij}$, and the segment joins $\mathbf{x}_i(t)$ to the closest of them, at delay $l = +120$ s. At the same instant j is about 1 km away: there is no simultaneous proximity. Top right: $\mathcal{M}_{ij}$ (green) in the (t, t') plane, inside the band $|t' - t| \le \delta$ (grey); the black segment is the column of the instant of the left panel. $\mathcal{M}_{ij}$ lies above the diagonal because j passes later. Bottom right: m_{ij} projects $\mathcal{M}_{ij}$ on the t axis, m_{ji} on the t' axis, and a_{ij} is their union: i is on the path of j from 90 to 775 s, j on the path of i from 220 to 885 s. Model: i flies at 8 m/s along $y = 600 \sin(2\pi s / 8\,\mathrm{km})$ m, where s is the distance flown; j joins the route at 300 s and leaves it at 800 s, drifting sideways at 3 m/s; altitude and heading conditions hold throughout.

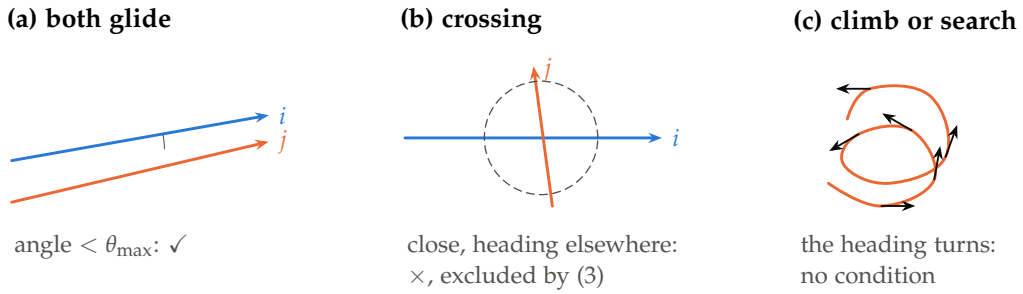


Figure 8: The heading condition (3). (a) Two gliders on the same route: their headings differ by less than $\theta_{\max}$. (b) A glider that crosses the area of i heading elsewhere is close, and the condition excludes it. (c) In a climb or a search the heading turns continuously and says nothing about the route: the condition is off.

4.3 The role of each term

Delay. With $t' = t$ the condition is simultaneous proximity; with $t' \neq t$ it recognises similar but shifted tracks, the case of (R2)–(R3). If $t' > t$, j passes where i was, later: j follows i. In Figure 7 all of $\mathcal{M}_{ij}$ lies above the diagonal. The contact asks only that some delay exists; how many seconds j is behind i is another question, taken up in Section 4.5.

Tolerances: a box, not a ball. ε_h translates the "nearby places" of (R3): the tracks need not pass through the same points. Height gets a tolerance of its own because a group is flat: horizontally it spreads over scales of a kilometre, vertically it overlaps within a few hundred metres. A single ε in $\|\Delta\mathbf{x}\|_{3D} < \varepsilon$ either makes altitude irrelevant, if it is of the order of a kilometre, or penalises the normal horizontal spread in glide, if it is of the order of 300 m (Figure 9). In a thermal the height is the only condition that acts: the pilots circle over the same spot, with a typical spiral radius of 20–26 m [1], so the horizontal distance stays far below ε_h, and only ε_z separates pilots circling at different heights. Since j climbs during the delay, the separation allowed there is ε_z plus the height j gains in δ: $|z_i(t) - z_j(t)| < \varepsilon_z + w\delta$, 540 m at a climb rate $w = 2$ m/s with $\delta = 2$ min.

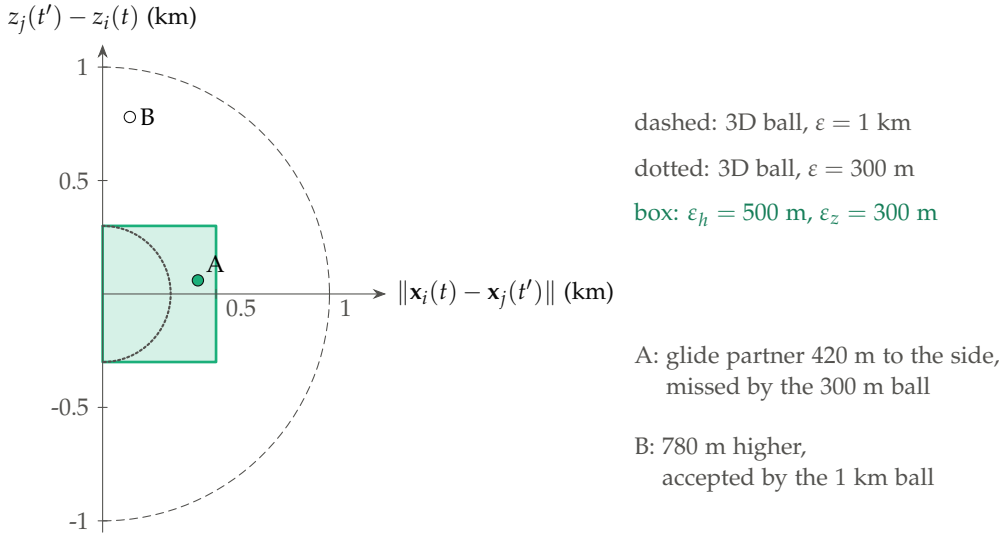


Figure 9: Tolerance regions for the separation between $(\mathbf{x}_i(t), z_i(t))$ and $(\mathbf{x}_j(t'), z_j(t'))$. The box of (2) is wide and flat, like a group: it keeps A and rejects B. A single 3D radius either rejects A or accepts B.

Heading only in glide. In glide the path is nearly straight and $\hat{\mathbf{v}}$ is the direction of the route: condition (3) excludes a pilot who crosses the same area heading elsewhere (Figure 8). In a climb the heading turns by 2π in about twenty seconds and carries no information on the route. In a search the pilot turns often, with partial circles and changes of heading, and the instantaneous heading says little about the route too: the condition is off whenever one of the two pilots is not in transition, unclassified fixes included. A rejected alternative is the velocity averaged over a window of a few minutes, with a minimum speed: in a long thermal without wind the mean velocity is nearly zero, and the threshold would remove the contacts exactly while two pilots share the thermal.

4.4 Choice of the maximum delay

If j retraces the track of i with a constant delay ℓ, $\mathbf{x}_j(t) = \mathbf{x}_i(t-\ell)$, the distance between the two pilots at the same instant is the distance between two points of the track of i flown ℓ apart,

$$\|\mathbf{x}_i(t) - \mathbf{x}_j(t)\| = \|\mathbf{x}_i(t) - \mathbf{x}_i(t-\ell)\| \leq v_{\max}\,\ell,$$

and $v_{\mathrm{xc}}\,\ell$ on average once ℓ is longer than a climb–glide cycle, v_{xc} being the mean speed of advance. The distance at the same instant grows with the delay. Requiring that two pilots shifted by any delay up to δ stay within visual range means $v_{\mathrm{xc}}\,\delta \leq \varepsilon_{\mathrm{vis}}$ (Figure 10):

$$\delta \lesssim \frac{\varepsilon_{\mathrm{vis}}}{v_{\mathrm{xc}}}, \tag{5}$$

so $\delta \approx 5$ min for $\varepsilon_{\mathrm{vis}} \approx 1.5$ km and $v_{\mathrm{xc}} \approx 5$ m/s. The speed v_{xc} must be measured on the data, for example as the daily median of the speed of advance over 30-minute windows. The bound gives the largest admissible delay, not the one at which following ends; on the data that one was measured directly and is 2 min (Section 8.1).

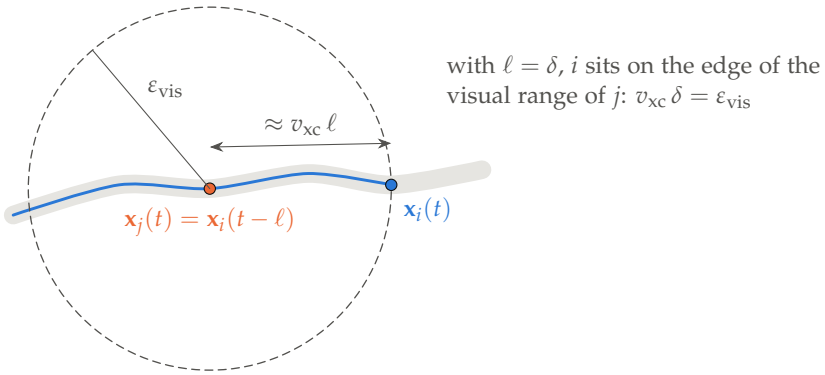


Figure 10: Choice of the maximum delay. A pilot that retraces the route of i with delay ℓ is about $v_{\mathrm{xc}}\ell$ behind; the dashed circle is the visual range around j. The largest delay that keeps i in sight is $\delta = \varepsilon_{\mathrm{vis}}/v_{\mathrm{xc}}$.

Remark 4.3 (two notions of "together"). Flying together may mean *seeing each other*, and so being able to use social information, or *exploiting the same air masses*. The two notions can be written as indicators:

$$\begin{aligned} \text{visual co-presence:}\quad & V_{ij}(t) = \mathbf{1}\Big[\big\|\big(\mathbf{x}_i(t), z_i(t)\big) - \big(\mathbf{x}_j(t), z_j(t)\big)\big\| < \varepsilon_{\mathrm{vis}}\Big], \\ \text{shared air:}\quad & m_{ij}(t) = \mathbf{1}\big[\exists\, |l| \leq \delta :\ \|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\| < \varepsilon_h, \\ & \qquad\qquad |z_i(t) - z_j(t+l)| < \varepsilon_z,\ D_{ij}(t, t+l) = 1\big]. \end{aligned}$$

Visual contact depends on the 3D distance and is simultaneous; exploiting the same thermals needs tight tolerances, vertical ones included, and admits a delay. Definition 4.2 with ε_z of a few hundred metres is the second notion, restricted to the visual range by (5). Widening ε_z brings the criterion closer to visual contact alone. The value of ε_z is therefore a choice of definition, to be declared (Figure 11).

4.5 How far behind: the delay of a contact

The contact asks only that some delay $|l| \leq \delta$ exists, and the groups of Section 5 are built from it alone. To say who follows whom, for the comparison between followers and leaders

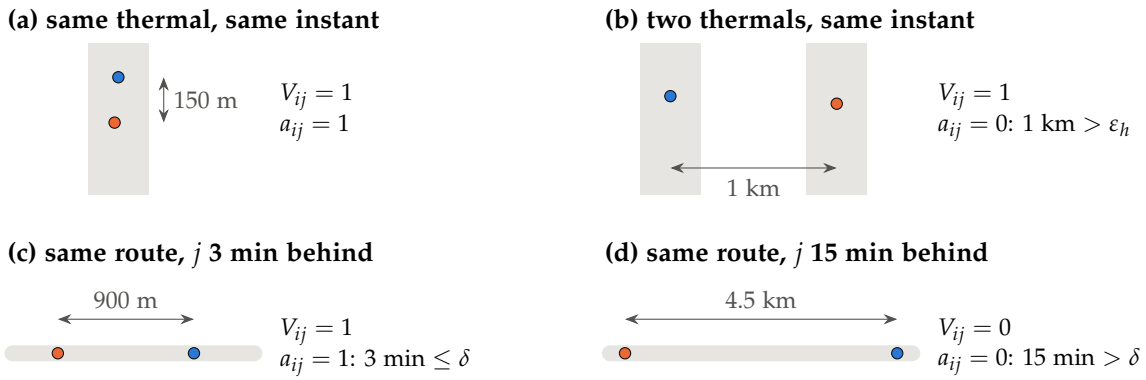


Figure 11: Visual co-presence V_{ij} and contact a_{ij} in four situations, with $\varepsilon_{\text{vis}} = 1.5$ km, $\delta = 5$ min and $v_{\text{xc}} = 5$ m/s. In (d), a delay of 15 minutes would put the two pilots in contact although they cannot see each other: this is what the bound (5) prevents.

(Section 11), we also need its value: how many seconds j is behind i. As in [15], at each instant t with $m_{ij}(t) = 1$ we take the delay at which j passes closest to where i is,

$$l^*(t) = \arg \min_{l:\,(t,t+l)\in\mathcal{M}_{ij}} \|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|, \tag{6}$$

and over a pair interval $[t_{\text{on}}, t_{\text{off}})$ (Definition 5.1) the signed median delay

$$\ell_{\text{med}} = \text{median}\{l^*(t) :\ t \in [t_{\text{on}}, t_{\text{off}}),\ m_{ij}(t) = 1\},$$

with $\ell_{\text{med}} > 0$ when j follows i. How well l^* is defined depends on the phase of flight.

Glide. The path is nearly straight and the distance has a single sharp minimum: l^* is the delay at which j crosses the point of i, $+120$ s in Figure 7.

Thermal. j circles over the same spot, and the distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$, as a function of l, has a nearly equal minimum at every turn (Figure 12, top right): the minimiser is degenerate, and $l^*(t)$ falls anywhere among the accepted delays. Altitude narrows them. During the climb at rate w, pilot j gains wl metres in l seconds:

$$z_j(t+l) \simeq z_j(t) + w\,l. \tag{7}$$

Write $\Delta z = z_i(t) - z_j(t)$ for how far j is below i at time t. The vertical condition then reads

$$|z_i(t) - z_j(t+l)| < \varepsilon_z \iff |\Delta z - w\,l| < \varepsilon_z \iff \Delta z - \varepsilon_z < w\,l < \Delta z + \varepsilon_z,$$

that is, dividing by $w > 0$,

$$l \in \left(\frac{\Delta z - \varepsilon_z}{w},\ \frac{\Delta z + \varepsilon_z}{w} \right), \tag{8}$$

an interval of width $2\varepsilon_z/w$ around $\Delta z/w$, the time j needs to reach the altitude of i; the scan cuts it to $[-\delta, \delta]$. With the starting $\delta = 5$ min this at least halves the range of l^*. With the chosen $\delta = 2$ min the interval, 300 s wide at $w = 2$ m/s, is wider than $[-\delta, \delta]$ and altitude gains little: at $\Delta z = 200$ m, from $[-120, 120]$ to $(-50, 120]$ s.

Search. The pilot turns without climbing, $w \simeq 0$, and altitude constrains no delay. The instants at which either pilot is in search are left out of the median; ℓ_{med} is undefined when none remains.

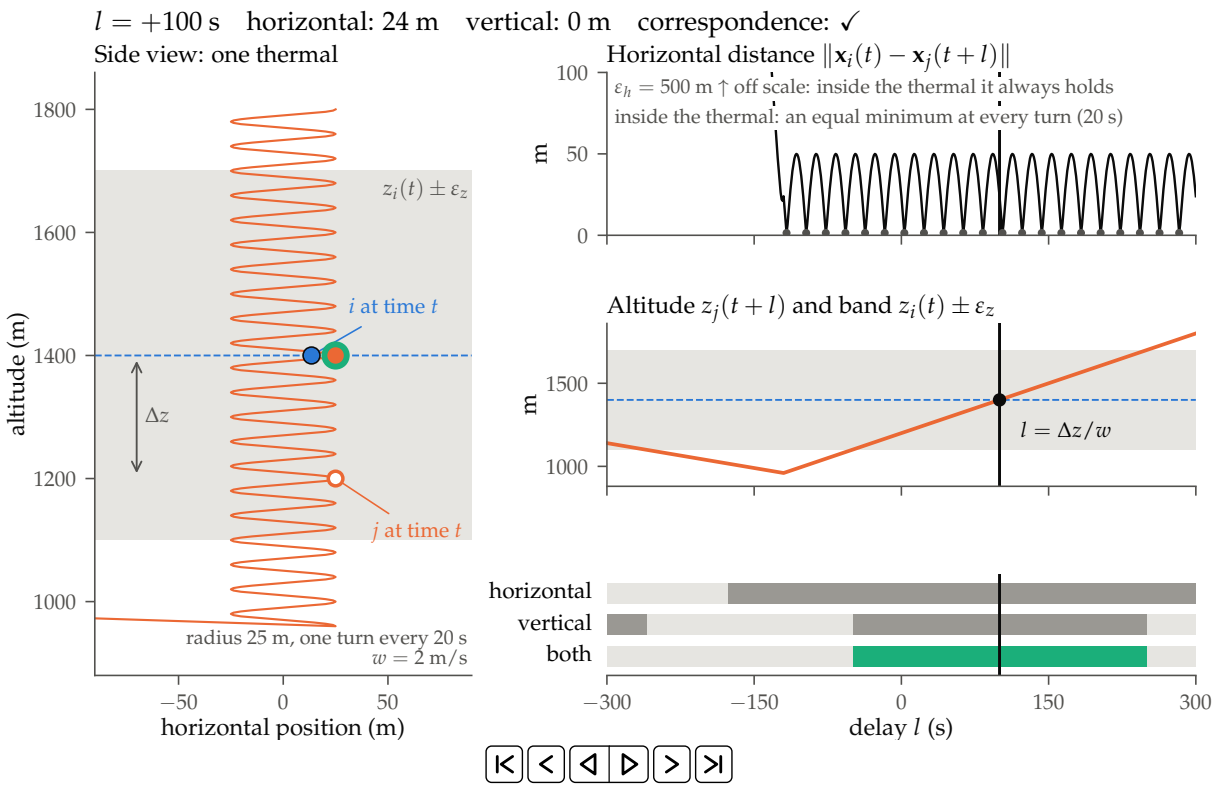


Figure 12: ▶ The delay in a thermal, for one instant t and every delay l; the animation runs the delay l from -300 to 300 s. Left: i and j in the same thermal, j lower by $\Delta z = 200$ m; the band is $z_i(t) \pm \varepsilon_z$. Top right: the horizontal distance between $\mathbf{x}_i(t)$ and $\mathbf{x}_j(t + l)$; inside the thermal it never exceeds 50 m, far below $\varepsilon_h = 500$ m, so the horizontal condition alone accepts every delay of the stay; it has an equal minimum at every turn, so it cannot fix l^*. Middle right: the altitude of j at $t + l$ crosses the band in a single window. Bottom: the delays that satisfy each condition and both; both hold for $-50 < l < 250$ s, the interval (8), centred on $\Delta z/w = 100$ s, and the scan keeps those with $|l| \le \delta$: $-50 < l \le 120$ s with $\delta = 2$ min. Before $l = -120$ s, j was gliding towards the thermal: at $l \approx -280$ s it had the right altitude but was 1.4 km away. Model: j circles with radius 25 m, one turn every 20 s, climbing at $w = 2$ m/s; it arrives at 9 m/s, sinking at 1 m/s; i is on the same circle.

5 From pairs to groups

5.1 Pair intervals

The instantaneous contact flickers wherever one of the quantities sits close to its threshold: a horizontal distance near ε_h in glide, a height difference near ε_z, an angle near $\theta_{\max}$. It also flickers at the changes of state, when the heading condition switches on and off, and wherever the phase labelling is wrong on a few fixes.

Definition 5.1 (pair interval)**.** For each pair consider the maximal runs of consecutive grid instants with $a_{ij} = 1$. Two runs separated by a gap shorter than g are joined; the resulting runs shorter than τ^* are dropped. Each remaining run is a half-open interval $[t_{\text{on}}, t_{\text{off}})$, and the set of the intervals of the day is $\mathcal{I}$.

Joining the short gaps plays the role of the robustness of Buchin et al. [6, §4], which ignores interruptions shorter than α; the maximal segments of Li et al. [15, 20] are an alternative. The threshold τ^* is fixed by the null model (Section 7).

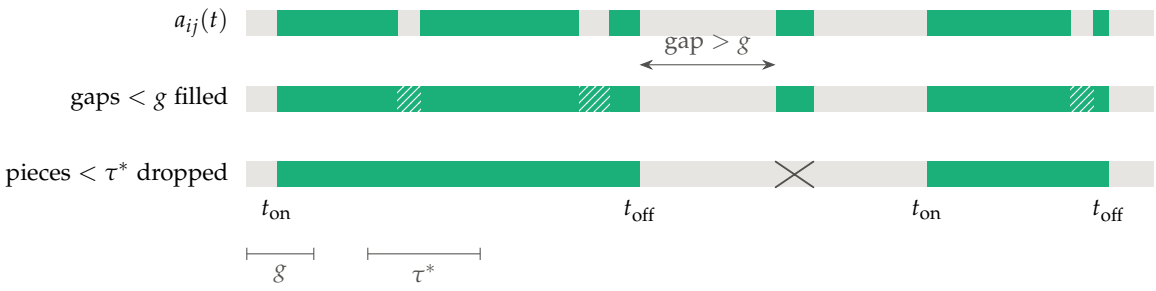


Figure 13: From the contact to the pair intervals. Top: the contact of one pair, with the flicker of the instantaneous condition. Middle: gaps shorter than g are filled (hatched). Bottom: runs shorter than τ^* are dropped; two intervals remain.

5.2 The problem of non-transitivity

Remark 5.2. The relation "i and j are in contact" is symmetric but not transitive. If 1 flies with 8 in the first part of the afternoon and 8 flies with 14 later, 1 and 14 may never have met.

A partition into groups presupposes an *equivalence relation*, a relation $\sim$ that is reflexive ($i \sim i$), symmetric ($i \sim j \Rightarrow j \sim i$) and transitive ($i \sim j$ and $j \sim k \Rightarrow i \sim k$). Contact lacks the third property, so any partition must give up something (Figure 14):

(a) direct contact between all members: taking the transitive closure puts 1 and 14 in the same group although they never met;

(b) the partition: groups may overlap, here $\{1, 8\}$ and $\{8, 14\}$;

(c) the static picture: the groups are defined instant by instant.

We adopt (c): at every instant the groups are the connected components of the graph of active contacts, as the components of Buchin et al. [6], and the evolution in time is described without giving the groups a persistent identity. Their maximal groups, which may overlap, are a form of (b).

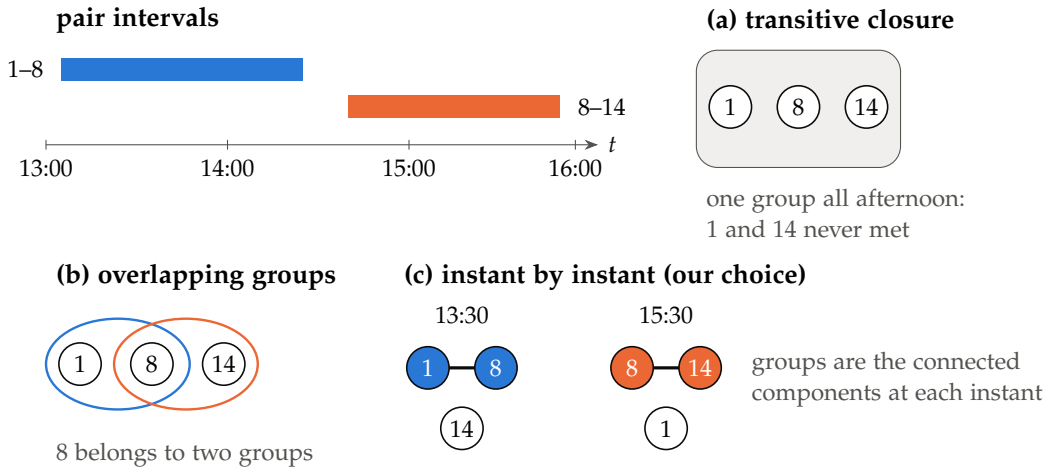


Figure 14: Non-transitivity. Flights 1 and 8 are in contact early in the afternoon, 8 and 14 later; 1 and 14 never meet. The three ways out: (a) one group containing all three, (b) two overlapping groups, (c) the groups of each instant, adopted here.

5.3 Configurations

The active graph at instant t is

$$G(t) = \big(V(t), E(t)\big), \qquad V(t) = \{i : \ t \in T_i^{\text{dep}}\},$$
$$E(t) = \big\{\{i,j\} : \ \exists\,(i,j,t_{\text{on}},t_{\text{off}}) \in \mathcal{I},\ t \in [t_{\text{on}},t_{\text{off}})\big\}. \tag{9}$$

In words. $V(t)$ are the pilots that at instant t have already departed, have not landed yet and have a position; $E(t)$ are the pairs whose contact interval contains t. A connected component of $G(t)$ is a group at instant t: its pilots are joined to each other by chains of contacts.

Definition 5.3 (configuration)**.** A configuration c is given by a set of flights M_c, its members, with $|M_c| \geq 2$, and by an interval of time $[t_c^{\text{start}}, t_c^{\text{end}})$, such that M_c is a connected component of $G(t)$ at every t of the interval and the interval is maximal. We write $c = \big(M_c, [t_c^{\text{start}}, t_c^{\text{end}})\big)$.

Inside a configuration the members are fixed.

Any change of members, one pilot joining or one leaving, ends the configuration and starts another. What can change inside a configuration is only the set of edges among its members.

The configurations are the edges of the Reeb graph of [6] with at least two entities, built on the graph of delayed contacts instead of simultaneous proximity. An edge that appears or disappears inside a component without changing its members, for example closing a triangle, does not create a new configuration (Figure 15).

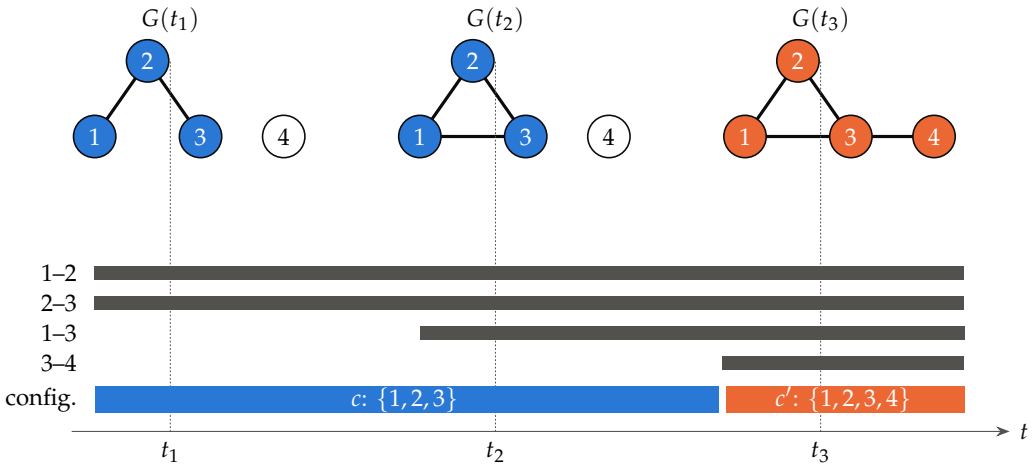


Figure 15: Configurations. Top: the active graph at three instants. Bottom: the pair intervals of the four flights and the configurations they produce. At t_2 the edge 1–3 closes a triangle: the component keeps its members and c goes on. When 4 joins, c ends and c' starts.

Proposition 5.4. *At every instant each flight belongs to at most one configuration.*

Proof. The connected components of $G(t)$ form a partition of $V(t)$. □

The membership of flight i at instant t is therefore well defined:

$$\gamma_i(t) = \begin{cases} \mathsf{L} & \text{if } k_i^0 \leq t < k_i^{\text{dep}} \text{ (local phase), or the flight is local,} \\ c & \text{if } i \in M_c \text{ and } t \in [t_c^{\text{start}}, t_c^{\text{end}}), \\ \mathsf{S} & \text{if } t \in T_i^{\text{dep}} \text{ otherwise (solo flight).} \end{cases} \tag{10}$$

In the tables, every instant between the departure and the landing that is not in a configuration is labelled S, including the instants in a gap of the track, where the flight has no position. The analyses that read the fixes see nothing there, so only the time fractions computed from the membership include them.

5.4 Episodes

Definition 5.5 (episode)**.** Two configurations are linked, $c \to c'$, if one starts exactly when the other ends and they have at least one member in common:

$$c \to c' \iff t_{c'}^{\text{start}} = t_c^{\text{end}} \ \wedge \ M_c \cap M_{c'} \neq \emptyset. \tag{11}$$

An episode is a connected component of the undirected graph associated with $\to$.

> **In words.** An episode is a chain of configurations without interruption: each one starts at the instant the previous one ends and keeps at least one of its members. Two configurations of the same episode need not share anyone; the chain joins them.

In the language of [6], an episode is a connected component of the Reeb graph after removing the edges of single pilots. Four properties matter (Figure 16).

(i) A split produces two configurations linked to the same previous one: they stay in the same episode. The same holds for a merge.

(ii) If the group dissolves completely, the chain breaks and the episode ends; a group that forms again later opens a new episode.

(iii) No threshold is needed: one member in common is enough.

(iv) The same flight can belong to several episodes of the same day, at different times.

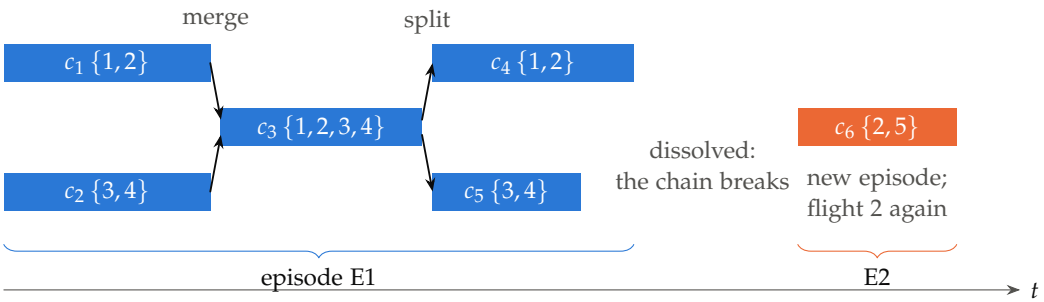


Figure 16: Episodes. Arrows are the links (11). The merge of c_1 and c_2 into c_3 and the split of c_3 into c_4 and c_5 keep all five configurations in episode E1, although c_1 and c_5 share no member. After the group dissolves, c_6 has no predecessor and opens E2: flight 2 belongs to both episodes.

Remark 5.6 (why not a persistent group name). One could keep the same group name when the overlap of the members between two instants exceeds a threshold, as methods that track communities in evolving networks do with the Jaccard index [9]. The choice has three defects: the threshold is arbitrary; if the members are replaced one at a time the group keeps its name even when none of the initial members is left; in splits and merges there is no natural criterion to decide who inherits the name. The episode does not claim to be "the same group": it records only the continuity, while the configuration records the composition.

5.5 Dependency blocks

Definition 5.7 (block)**.** For each day consider the bipartite graph flights–episodes, with an edge between a flight and every episode it belongs to. Its connected components, restricted to the flights, are the *dependency blocks*; a flight never in a group is a block on its own.

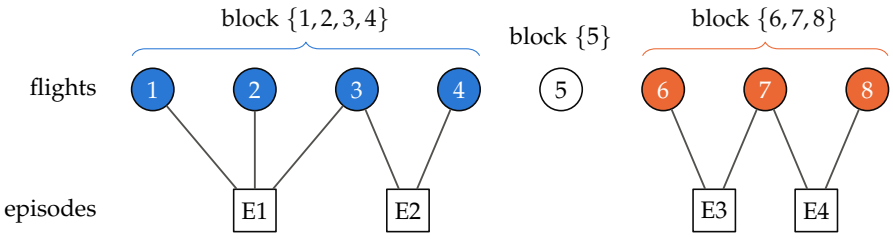


Figure 17: Dependency blocks of a day. Flight 3 is in episodes E1 and E2, so flights 1–4 form one block although 1 and 4 share no episode; flight 5 never flies in a group.

Proposition 5.8. *The blocks are the finest partition of the flights in which the members of every episode are in the same block.*

Proof. A partition with this property must contain the transitive closure of the relation "share an episode", whose classes are exactly the connected components of the bipartite graph; these, in turn, satisfy the property. □

The blocks are the units resampled in the bootstrap (Section 11): flights of different blocks never shared an episode, flights of the same block did. Buchin et al. already note that a grouping structure splits the trajectories into independent data sets [6, §1].

5.6 An example day

Figure 18 follows a day with five flights. At 11:40 flights 1, 8 and 22 meet (edges 1–8 and 8–22): configuration $c_1 = \{1, 8, 22\}$ starts. At 12:05 the edge 1–22 closes the triangle: the members do not change and c_1 goes on. At 12:55 the edge 1–8 closes and 1 changes route: c_1 ends and $c_2 = \{8, 22\}$ starts; they share 8 and 22, so $c_1 \to c_2$. At 13:30 the edge 8–14 opens: $c_3 = \{8, 14, 22\}$, and $c_2 \to c_3$. At 14:10 the group dissolves and episode E1 ends. At 15:00 flights 8 and 22 meet again: $c_4 = \{8, 22\}$ has no predecessor and opens episode E2.

Flights 1 and 14 belong to the same episode without ever being in the same configuration; 8 and 22 belong to two episodes; 1, 8, 14 and 22 form one dependency block, and flight 5, never in a group, a block of its own.

6 Algorithm

The definitions of Sections 4 and 5 can be computed literally: for every pair of flights of a day, for every instant of one and every delay up to δ, test the four conditions of (2). On a day with 200 flights in the air this is $200 \cdot 199/2 = 19\,900$ pairs; with 1.5 h of flight on the 5-s grid, 1 080 instants, and the 49 delays of $\delta = 2$ min, it is about 10^9 tests for that day alone. Almost all of them are wasted: most pairs of the day are never within several kilometres of each other. The algorithm therefore works in phases (Figure 19). Phase 0 puts every flight on a common time grid; Phase 1 discards, with a cheap test that provably loses nothing, the pairs and the instants that cannot have a correspondence; Phase 2 applies the exact test to what remains; Phase 3 turns the contact into intervals; Phase 4 builds the groups from the intervals with a sweep over time.

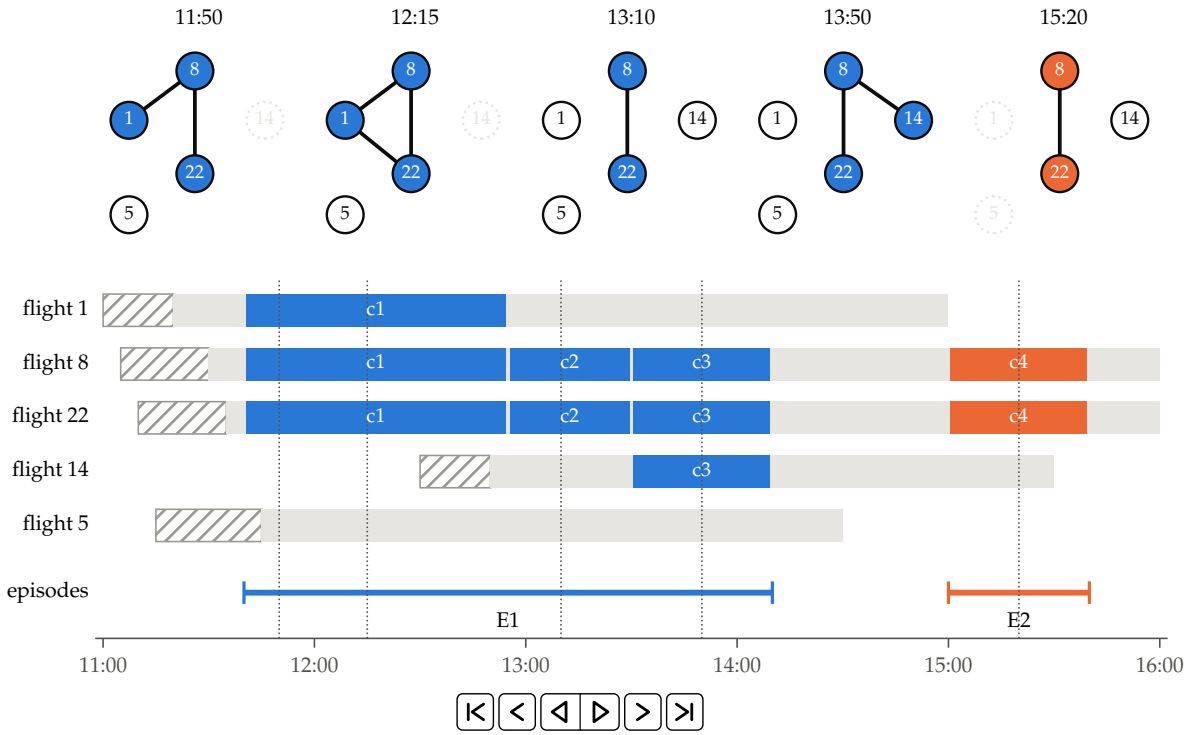


Figure 18: ▶ An example day; the animation runs the day from 11:00 to 16:00 and stops at each event. Top: the active graph $G(t)$ at the five dotted instants; filled nodes are in a configuration, coloured by episode; dashed nodes are in the local phase, faint ones not in flight. Bottom: the membership $\gamma_i(t)$ of each flight: hatched is the local phase L, grey is solo S, coloured segments are the configurations c_1–c_4, and the last row spans the episodes.

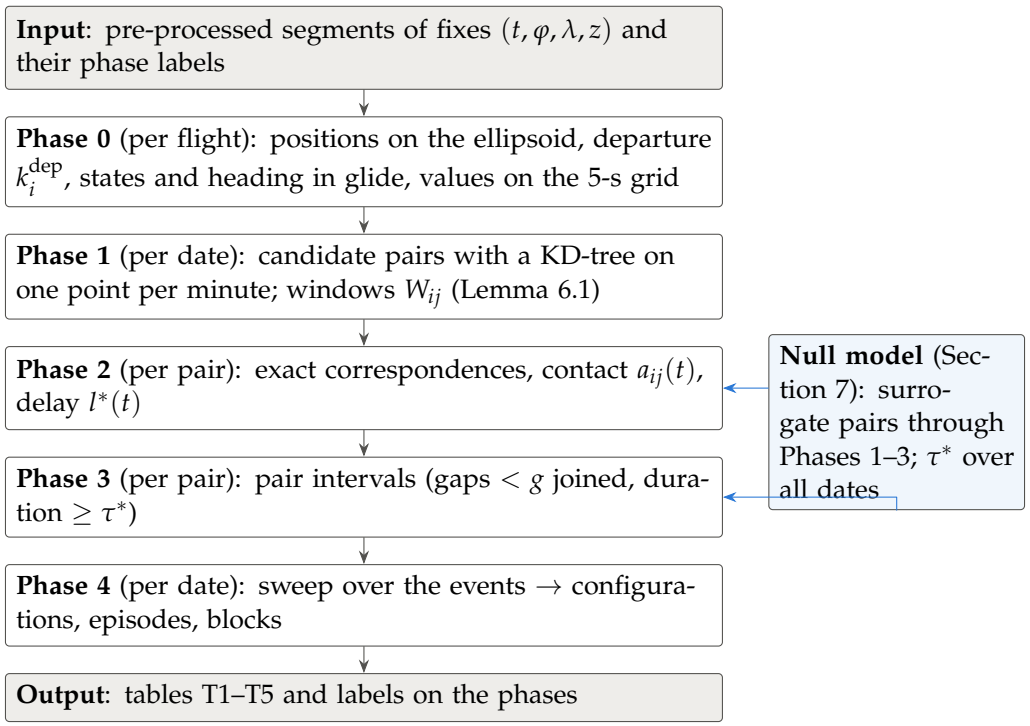


Figure 19: The method as a pipeline. Phases 0–3 and the null model run date by date; τ^* is computed once, over all dates; Phase 4 then runs date by date with the intervals lasting at least τ^*.

6.1 How a run is organised

Flights of different days never meet, so every phase but the calibration works on one date at a time: the dates run in parallel, and the memory needed is that of one day. A run has three stages.

1. **Per date.** Phase 0 puts the flights of the date on the grid; Phases 1–3 find the contact intervals of the real pairs, with their duration and ℓ_{med}, but without the threshold τ^*, which is not known yet. The null model of Section 7.1 draws the surrogate pairs of the date and sends each one through the same Phases 1–3, with the copy of j shifted in time and filtered against i alone. The result of each date is saved, so an interrupted run resumes.

2. **Once.** τ^* follows from the real and the surrogate intervals of all the dates, Eq. (14).

3. **Per date.** Phase 4 keeps the intervals lasting at least τ^* and builds configurations, episodes and blocks.

On the whole archive, 6 209 dates, a run takes about an hour with five processes.

6.2 Phase 0: preparation per flight

(a) **Positions.** The Earth-centred position of every fix (Section 3). One frame serves all the flights, at any distance and without a map projection, so the coordinates of two flights compare directly.

(b) **Altitude.** GNSS z, unchanged.

(c) **Departure.** k_i^{dep} from (1).

(d) **States and heading.** $s_i(k)$ from the phase labelling. In glide, $\hat{\mathbf{v}}_i(k)$ is the direction of the horizontal velocity, filtered with Savitzky–Golay, in the east and north directions of the first fix of the flight (the frame of the cleaned tracks), not of the fix itself. Two flights are then compared in frames whose axes differ by the rotation of the local north between their first fixes and the points compared: for a displacement d to the east it is $(d/R)\tan\varphi$, about 3° for 300 km at latitude 45°, small against $\theta_{\max} = 45°$.

(e) **Grid.** The instants of $\Delta\mathbb{Z}$, with $\Delta = 5$ s, the same for all flights. A grid instant takes the position and the altitude of the fix at that instant or, between two fixes of the same segment at most $h_{\max} = 10$ s apart, the values interpolated linearly between them, with the state and the heading of the nearer fix; elsewhere the flight has no position. On a logger that records every second every grid instant is a fix. An interpolated position is at most $v_{\max}h/2$ from the true one, for two fixes h apart: 69 m at 5 s, 139 m at 10 s.

Is $\Delta = 5$ s enough in a thermal?

Four points per turn do not draw the spiral (Figure 20), but the contact does not need its shape. What matters is how far a grid position can be from the track between two grid instants, compared with the tolerances:

- horizontally, a pilot is never more than $v_{\max}\Delta/2 \approx 69$ m from the nearest grid point in the worst case, about 20 m while circling at 8 m/s; $\varepsilon_h = 500$ m, and a whole thermal of radius 25 m fits in a small fraction of it;
- vertically, $w\Delta/2 = 5$ m at 2 m/s and at most 12.5 m at 5 m/s, against $\varepsilon_z = 300$ m;

- on the delay, a grid of 5 s against $\delta = 120$ s; in a thermal l^* is undetermined over the whole window of accepted delays anyway (Section 4.5);
- on the ends of the intervals, $\pm\Delta$ against $g = 120$ s.

States and headings are computed on the fixes, before the grid, and lose nothing. The check of Section 12.1 repeats the computation with $\Delta = 1$ s on a subset of 1-Hz pairs.

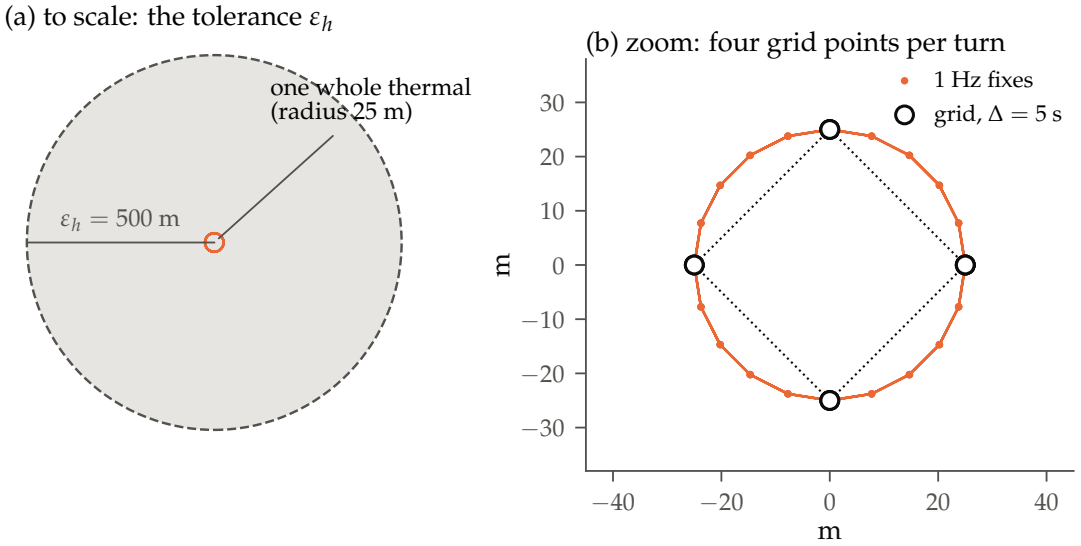


Figure 20: The 5-s grid in a thermal, to scale. (a) The horizontal tolerance ε_h and a whole thermal of radius 25 m. (b) One turn of 20 s sampled at 1 Hz and on the grid of $\Delta = 5$ s: four grid points per turn, each within 20 m of the 1 Hz fixes around it.

`date`	`flight`	`t`	`X, Y, Z`	`z`	`ve, vn`	`state`
partition	int32	int32 (s)	float32	float32	float16	int8

Table 4: Table of the grid points, in Parquet partitioned by date: about 30 bytes per row. `X, Y, Z` are the Earth-centred position, to 0.5 m in float32; `ve, vn` the unit heading in the east and north of the point.

Under the assumption, to be checked, of 1.86×10^5 flights lasting 1.5 h on average, the table has about 2×10^8 rows: 6 GB uncompressed, less in Parquet. It is worth keeping, because the null model and the calibration read it several times.

6.3 Phase 1: candidate pairs

Phase 1 answers one question for each date: which pairs of flights, and at which instants, can have a correspondence at all? It does so with a necessary condition that is cheap to test. If i at t is within ε_h of j at t', with $|t' - t| \leq \delta$, then at the same instant the two pilots cannot be farther apart than ε_h plus the distance j can fly in δ. Testing this distance at every instant would cost as much as the exact test, so it is tested only at whole minutes, with a radius widened to cover the rounding.

Pieces of track. The bound needs a top speed: between any two instants of a piece of track, the pilot moves at most $v_{\max}$ times the time between them, with $v_{\max} = 100$ km/h, the top speed of the pre-processing. A *piece* is a run of grid positions of one segment of one flight with

no step faster than $v_{\max}$; a faster step, which the excursion rule of the pre-processing should already have removed, ends a piece. Let $\tilde{\mathbf{x}}_i(s)$ be a piece of flight i extended by holding its first position before it and its last position after it. The filter reads positions at whole minutes, interpolated linearly between grid instants; a minute may fall just before or just after a piece, and the extension gives the track a position there.

Lemma 6.1 (no false negative). *If $(t, t') \in \mathcal{M}_{ij}$, $\tilde{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}_j$ are the extended pieces that contain t and t', and $t_c = 60 \cdot \mathrm{round}(t/60)$, then*

$$\|\tilde{\mathbf{x}}_i(t_c) - \tilde{\mathbf{x}}_j(t_c)\| \le D_c := \varepsilon_h + v_{\max}\,(\delta + 60\ \mathrm{s}). \tag{12}$$

> **In words.** t_c is the whole minute nearest to t (Figure 21a), and both positions in (12) are read at that *same* instant. If two pilots have a correspondence at all, then at the nearest whole minute they are less than D_c apart; a pair farther than D_c at every minute has no correspondence and can be discarded.

Proof. The extension with a fixed position keeps the Lipschitz property $\|\tilde{\mathbf{x}}(s) - \tilde{\mathbf{x}}(s')\| \le v_{\max}|s - s'|$ for all s, s'. By construction of t_c, $|t - t_c| \le 30$ s, and $|t' - t_c| \le |t' - t| + |t - t_c| \le \delta + 30$ s. The triangle inequality along the path $\tilde{\mathbf{x}}_i(t_c) \to \mathbf{x}_i(t) \to \mathbf{x}_j(t') \to \tilde{\mathbf{x}}_j(t_c)$ (Figure 21) gives

$$\begin{aligned}\|\tilde{\mathbf{x}}_i(t_c) - \tilde{\mathbf{x}}_j(t_c)\| &\le \|\tilde{\mathbf{x}}_i(t_c) - \mathbf{x}_i(t)\| + \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| + \|\mathbf{x}_j(t') - \tilde{\mathbf{x}}_j(t_c)\| \\ &\le 30\, v_{\max} + \varepsilon_h + v_{\max}(\delta + 30),\end{aligned}$$

where the middle term is below ε_h because (t, t') is a correspondence (Definition 4.1). □

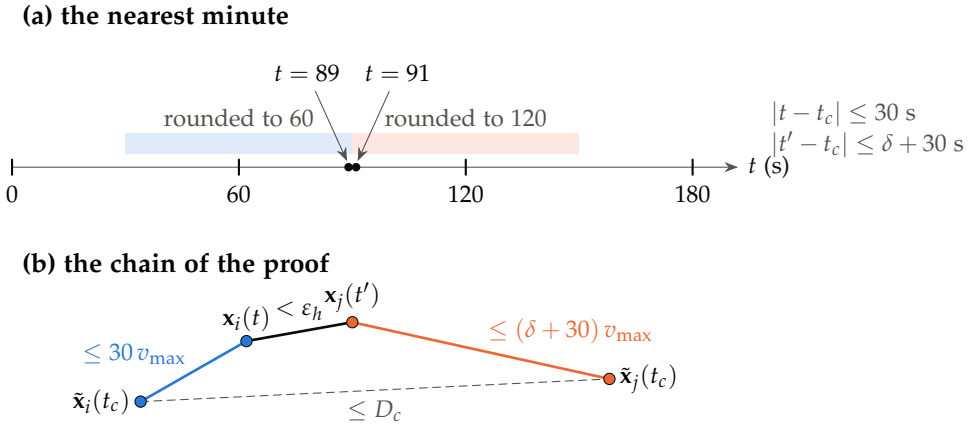


Figure 21: Lemma 6.1. (a) Every instant is at most 30 s from its nearest whole minute. (b) From i at the minute to j at the same minute through the correspondence (t, t'): each step is bounded by the time it takes at most $v_{\max}$, or by ε_h.

The radius. With $v_{\max} = 100$ km/h, $\delta = 300$ s and $\varepsilon_h = 500$ m, $D_c \approx 10.5$ km; with the delay chosen on the data, $\delta = 120$ s, $D_c \approx 5.5$ km. The spacing of one minute is a compromise: with points every s seconds the same proof gives $D_c = \varepsilon_h + v_{\max}(\delta + s)$, so a finer spacing narrows the radius a little but multiplies the points to compare.

What a KD-tree is. Phase 1 must find, among the points of one minute, all the pairs closer than D_c. Comparing every point with every other costs $n^2/2$ distances for n points. A KD-tree (a k-dimensional tree) avoids most of them. It splits the points in two at the median of one coordinate, then each half at the median of the next coordinate, and so on, until each cell holds a few points (Figure 22). Every node of the tree is a box of space, and the leaves partition the points. To find the points within D_c of a given point, the search starts at the root and descends only into the boxes that the ball of radius D_c reaches; a box farther than D_c from the point is skipped with all its points, which are never compared. Building the tree takes $O(n \log n)$ operations, and when the points are spread out a search visits $O(\log n)$ boxes plus those that touch the ball. The search of all pairs within D_c (`query_pairs` of `cKDTree` in SciPy [25]) runs this for every point and returns each pair once.

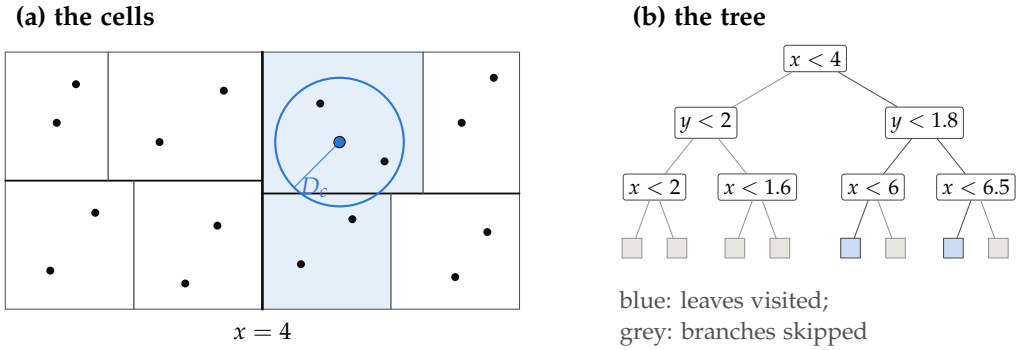


Figure 22: A KD-tree in two dimensions. (a) The sixteen points are split by a vertical line at the median of x, each half by a horizontal line at the median of its y, each quarter by a vertical line again: eight cells of two points. A search of radius D_c around the blue point visits only the two cells that the circle reaches (shaded) and compares their four points; the other twelve are never looked at. (b) The same splits as a tree: the search drops the left branch at the root, since the whole circle lies at $x > 4$, and further down every cell that the circle does not reach.

One tree for all the minutes. Phase 1 needs the pairs of points of the *same* minute only. Building one tree per minute would work; a single tree does the same job with a fourth coordinate. Every point gets the coordinates $(X, Y, Z, K\,t_c/60)$, its Earth-centred position and its minute multiplied by $K = 10^8$ m. Two points of the same minute have the same fourth coordinate, so their distance in four dimensions is their distance in space; two points of different minutes are at least $K \gg D_c$ apart and are never returned (Figure 23).

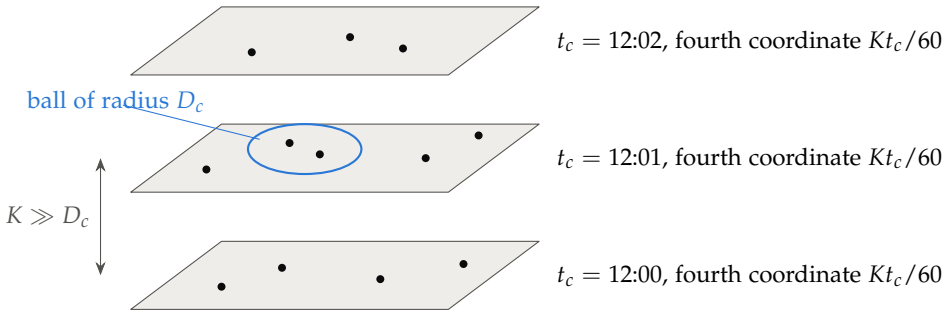


Figure 23: The fourth coordinate. Each minute is a layer of the space $(X, Y, Z, Kt_c/60)$, drawn here as a plane, and the layers are $K \gg D_c$ apart: a search of radius D_c only finds points of the same minute, so one tree does the work of one search per minute.

The steps, for each date.

1. For each piece, one point per whole minute of its extended track, for t_c from $\delta + 60$ s before its first instant to $\delta + 60$ s after its last. This is the interval in which the lemma may ask for the position, whichever of the two roles the flight has.
2. One KD-tree on the four coordinates of all the points of the date.
3. `query_pairs` with radius D_c: pairs of points of the same minute, hence triples (i, j, t_c); the pairs of two pieces of the same flight are dropped.
4. For each candidate pair, with $i < j$, the window is $W_{ij} = \bigcup_{t_c} [t_c - 30, t_c + 30] \cap T_i^{\text{dep}}$, the union over the minutes t_c at which the pair passed the filter: a set of grid instants of the flight of i.

By the lemma, every correspondence $(t, t') \in \mathcal{M}_{ij}$ has $t \in W_{ij}$, and every pair with a correspondence is a candidate. The cost is $O(n_d \log n_d)$ per date, with n_d the number of points of the day. Phase 1 is worth running on the whole dataset before Phase 2: the number of pairs and the total duration of the windows give a precise estimate of the cost that follows.

6.4 Phase 2: exact correspondences

For one candidate pair, Phase 2 tests Definition 4.1 exactly, at every instant t of the window W_{ij} and every delay l of the grid up to δ, and keeps three results: $m_{ij}(t)$, whether i at t has a correspondence; $m_{ji}(t')$, whether j at t' has one; and $l^*(t)$, the delay of the closest correspondence of i at t (Section 4.5).

Lemma 6.2. *Scanning the instants $t \in W_{ij}$ and the delays $l \in \Lambda = \{-L\Delta, \dots, L\Delta\}$, with $L = \lfloor \delta/\Delta \rfloor$, enumerates all pairs of $\mathcal{M}_{ij}$; hence m_{ij} and m_{ji} are obtained exactly.*

Proof. Every $(t, t') \in \mathcal{M}_{ij}$ has $t \in W_{ij}$ (Lemma 6.1) and $t' - t \in \Lambda$. Moreover $m_{ji}(t') = 1$ if and only if there is a t with $(t, t') \in \mathcal{M}_{ij}$, and every such t lies in W_{ij}. □

Algorithm 1 Phase 2: scan of one candidate pair

Input: flights $i < j$, on the grid; window $W \subseteq T_i^{\text{dep}}$; delays Λ
Output: contact a on the support $W \cup (W + \Lambda)$; delay l^* on W

$m_{ij} \leftarrow 0$; $m_{ji} \leftarrow 0$; $d^{\min} \leftarrow +\infty$; $l^* \leftarrow$ undefined
for $l \in \Lambda$ **do**
  $\tau \leftarrow \{t \in W : \ t + l \in T_j^{\text{dep}}\}$ ▷ array of instants
  $d \leftarrow \|\mathbf{x}_i(\tau) - \mathbf{x}_j(\tau + l)\|$; $h \leftarrow |z_i(\tau) - z_j(\tau + l)|$
  $c \leftarrow (d < \varepsilon_h) \wedge (h < \varepsilon_z) \wedge D_{ij}(\tau, \tau + l)$
  $m_{ij}[\tau] \leftarrow m_{ij}[\tau] \vee c$; $m_{ji}[\tau + l] \leftarrow m_{ji}[\tau + l] \vee c$
  $u \leftarrow c \wedge (d < d^{\min}[\tau])$; $d^{\min}[\tau] \leftarrow d$ and $l^*[\tau] \leftarrow l$ where u
$a(t) \leftarrow m_{ij}(t) \vee m_{ji}(t)$
return a, l^*

In words. Each pass of the loop fixes one delay l and compares i at every instant of the window with j at the instant l later: one distance, one height difference and, if both glide, one angle per instant. Where the three conditions hold, i at t and j at $t + l$ both get a correspondence. A running minimum of the distance over the delays that pass gives $l^*(t)$.

In the (t, t') plane a pass is the diagonal $t' = t + l$, evaluated over the whole window at once (Figure 24). The code evaluates all the delays of up to 2 000 instants at once, as a matrix of instants by delays; the result is the same as the loop. The cost is $\sum_{\text{pairs}} |W_{ij}|\,(2L+1)$ distance evaluations, with $2L+1 = 121$ for $\delta = 5$ min and $\Delta = 5$ s, 49 for $\delta = 2$ min. The memory is $O(|W_{ij}|)$ per pair: boolean arrays are accumulated, without generating the list of pairs of points.

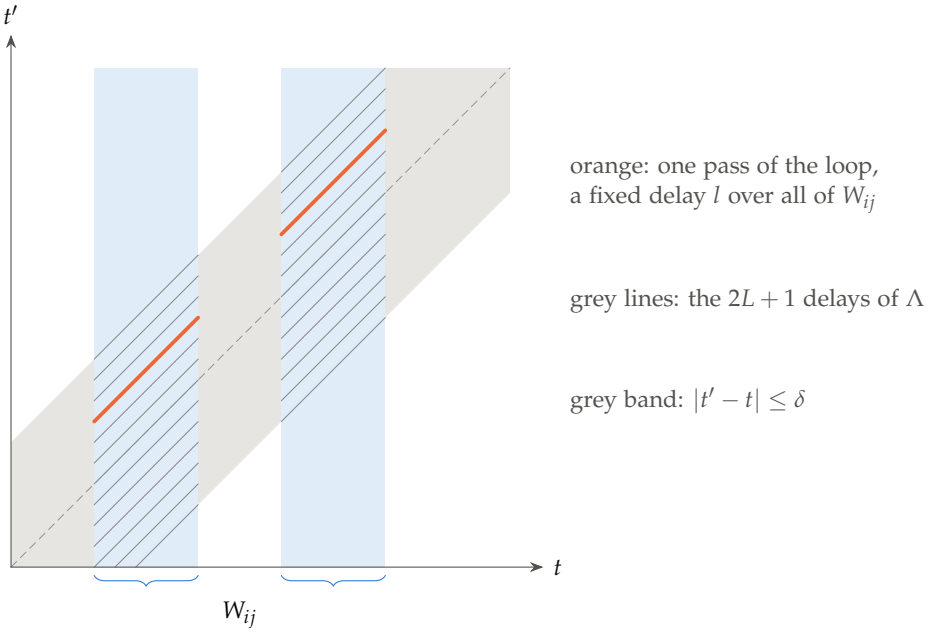


Figure 24: Phase 2 in the (t, t') plane. Only the columns of the window W_{ij} are scanned, one diagonal $t' = t + l$ at a time; by Lemma 6.2 this covers all of $\mathcal{M}_{ij}$.

6.5 Phase 3: pair intervals

Phase 3 turns the contact of a pair into the intervals of Definition 5.1.

1. The contact is restricted to the time from the later of the two departures to the earlier of the two landings, so that every edge of the graph (9) joins two nodes of $V(t)$. This cuts at most δ at each end of a contact, where m_{ji} could hold at an instant at which i has not yet departed or has already landed.

2. The instants with $a_{ij} = 1$, sorted, are cut wherever two consecutive ones are separated by a gap of at least g; each run becomes a half-open interval $[t_{\text{on}}, t_{\text{off}})$, with t_{off} one grid step after its last instant.

3. The intervals are kept with their duration. The threshold τ^* is applied after the calibration (Section 6.1), and only intervals lasting at least τ^* enter Phase 4.

For each interval one also records:

- the signed median delay ℓ_{med}, which tells who follows whom and does not enter Phase 4 (Section 4.5);
- the distance between the takeoffs $d_{\text{dec}} = \|\mathbf{x}_i(k_i^0) - \mathbf{x}_j(k_j^0)\|$, which allows requirement (R1) to be imposed afterwards, if wanted.

6.6 Phase 4: configurations and episodes

Phase 4 needs the components of the active graph $G(t)$ at every instant, but $G(t)$ changes only when an interval starts or ends. Between one event and the next the set of active edges is fixed, so the components are recomputed only at the events, and only around the nodes involved (Figure 25).

> **In words.** The events of the day, the starts and ends of the intervals, are sorted in time. At each event the touched flights and the open configurations they belong to form a set R; the components inside R are recomputed. An open configuration whose members no longer form a component is closed; a new component opens a configuration, linked by a union–find to every configuration just closed that shares a member with it. The classes of the union–find are the episodes.

Algorithm 2 Phase 4: sweep over the events of one day

Input: pair intervals $\mathcal{I}$ of the day
Output: configurations; episodes (classes of a union–find)

events $\leftarrow \{(t_{\text{on}}, +, \{a, b\}),\ (t_{\text{off}}, -, \{a, b\})\}$ sorted in time
$A \leftarrow \emptyset$ (active edges); $\mathcal{O} \leftarrow \emptyset$ (open configurations)
for each instant t_e, with its batch of events B **do**
  $U \leftarrow$ endpoints of the edges in B ▷ nodes involved
  $R \leftarrow U \cup \bigcup\{M_c :\ c \in \mathcal{O},\ M_c \cap U \neq \emptyset\}$
  update A with the events of B
  $\mathcal{K} \leftarrow$ connected components of $(R, A|_R)$ with at least two nodes ▷ local visit
  $X \leftarrow \{c \in \mathcal{O} :\ M_c \subseteq R,\ M_c \notin \mathcal{K}\}$; close every $c \in X$ with $t_c^{\text{end}} \leftarrow t_e$
  for $K \in \mathcal{K}$ that is not already an open configuration **do**
    open c' with $M_{c'} = K$ and $t_{c'}^{\text{start}} = t_e$
    for $c \in X$ with $M_c \cap K \neq \emptyset$ **do**
      UNION(c, c')

Lemma 6.3. *After the update, the connected component of every node of R is contained in R; hence the local recomputation of Algorithm 2 gives the same components as a global one.*

Proof. Before the update the open configurations are disjoint, and those that touch U lie entirely in R. A path that starts in R uses old edges, which stay inside the old component, or new edges, whose endpoints lie in $U \subseteq R$. By induction on the length of the path, every node it reaches is in R. □

The cost per day is $O(|\mathcal{I}| \log |\mathcal{I}|)$ for the sorting, plus the sum of the sizes of the components involved at each event, at most a few hundred nodes. The classes of the union–find [24] are the episodes. The dependency blocks follow from a second union–find, which joins the flights that share an episode; a flight never in a group is a block on its own. From the list of the configurations one obtains $\gamma_i(t)$ as in (10), filling with S the stretches after the departure not covered by a configuration and with L the local phase.

6.7 Costs and space

Phases 1–4 are independent across dates and run in parallel without coordination. Phase 2 dominates the cost; Phase 1 allows it to be estimated before running it.

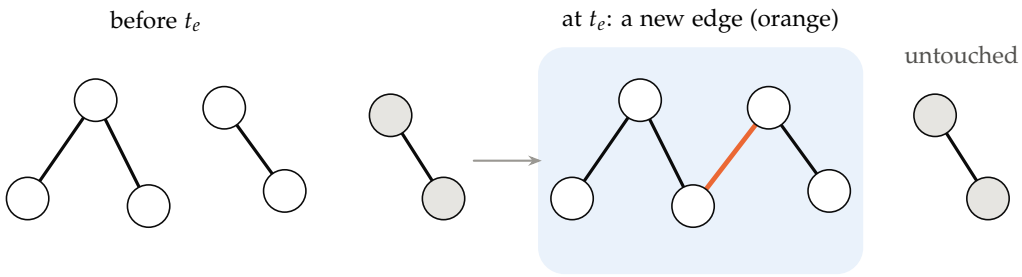


Figure 25: One step of the sweep. Only the nodes of R are visited again; the grey component far away keeps its configuration. The animation of Figure 18 runs the sweep over a whole day.

Phase	Cost	Data produced
0	linear in the number of fixes ($\sim 10^9$)	grid points, $\sim 2 \times 10^8$ rows
1	$O(n_d \log n_d)$ per date, $\sum_d n_d \sim 1.7 \times 10^7$	candidate pairs and windows
2	$\sum \lvert W_{ij} \rvert (2L+1)$; measured after Phase 1	temporary arrays per pair
3	linear in the duration of the windows	T1, pair intervals
4	$O(\lvert\mathcal{I}\rvert \log \lvert\mathcal{I}\rvert)$ per date	T2–T5

Table 5: Costs and space. The numbers assume 1.86×10^5 flights of 1.5 h on average and are to be checked on the data.

7 Calibration: what chance means

Two pilots who never interact still meet: they take off from the same site, follow the same ridges and valleys and climb in the same house thermals. A contact therefore says little by itself; what counts is how often contacts of a given duration occur beyond chance. Interaction indices based on standard tests overestimate the interactions with finely sampled telemetry [16], so chance is built from the data, with the null model of this section, and the minimum duration τ^* follows from it. The run reported here and in the next two sections covers the 162 485 cleaned flights of 6 209 dates of the FFVL archive (Section 3).

7.1 The null model

On each date, up to 200 pairs (i, j) of flights whose takeoffs are less than $R_s = 10$ km apart are drawn; in each, flight j is replaced by its copy shifted in time:

$$\mathbf{x}_j^s(t) = \mathbf{x}_j(t + r_{ij}), \quad z_j^s(t) = z_j(t + r_{ij}), \qquad r_{ij} = (k_j^0 - k_i^0) - \vartheta, \quad \vartheta \sim \hat{P}(k_j^0 - k_i^0). \tag{13}$$

The variable ϑ is drawn from the delays between the takeoffs of the pairs of that date with takeoffs within R_s, so the copy of j takes off ϑ after i, as a real pilot of the day could. Only shifts with $|r_{ij}| \geq r_{\min} = 2\delta + \Delta = 245$ s are accepted, rounded to the grid step. A pair enters once, with the roles of i and j drawn at random. The idea of shifting tracks in time, which keeps the spatial constraints and destroys the synchronisation, comes from Spiegel et al. [22], who shuffle whole days of each individual; here a flight is shifted within its day, against one other flight.

No real correspondence survives the shift. If (t, t') was a real correspondence, in the surrogate pair the position of j at instant t' appears at instant $t' - r_{ij}$, which is at least $|r_{ij}| - \delta > \delta$ away from t (Figure 26). The correspondences that remain are encounters on the same route without synchrony.

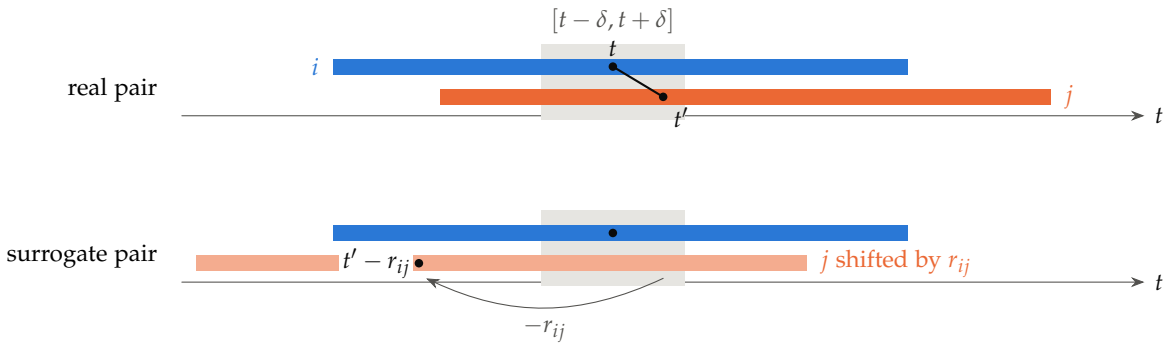


Figure 26: Null model. Top: a real correspondence (t, t'), with t' inside the window $[t-\delta, t+\delta]$ (grey). Bottom: the track of j shifted by r_{ij} with $|r_{ij}| > 2\delta$; the same point of j now comes at $t' - r_{ij}$, outside the window, so the real correspondence is gone and only chance encounters remain.

What the shifted pair keeps: the route, the terrain, the date, and the takeoff delays of the day as a distribution. What it destroys: every correspondence within δ, since $|r_{ij}| > 2\delta$. On the archive the shift is far larger than its minimum (Figure 27): 58 min in the median, and 90% of the shifts lie between 8 min and 4.4 h. So the copy also flies in the weather of another hour, with other thermals working and another cloud base.

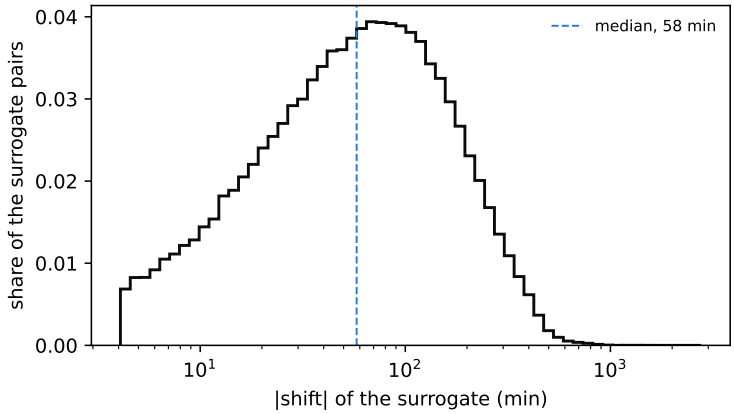


Figure 27: The time shifts $|r_{ij}|$ of the 248 741 surrogate pairs of the chosen run, whole archive.

Chance is therefore an operational notion: the contacts that the same routes give without synchrony. Two consequences follow.

- Pilots who use the same thermals at the same time without following each other are in synchrony with the weather of the hour, which the shift removes: their contacts count as real.
- A pilot who follows another deliberately, but more than δ behind, matches the other only beyond δ: the real pair never shows this contact, while a surrogate whose shift brings that delay within δ counts it as chance.

7.2 The false discovery rate and τ^*

The surrogate pairs go through Phases 1–3 like the real ones (Section 6.1), and their intervals are the contacts that chance alone gives. Since the surrogates are a sample of the pairs of each date, their rate is taken date by date. On each date d:

1. $\mathcal{P}_d$ is the set of the pairs (a, b), $a < b$, of flights of the date with takeoffs less than R_s apart.

2. The exposure of a pair is the time both fly after departure, $h_{ab} = \max\big(\min(k_a^{\text{fin}}, k_b^{\text{fin}}) - \max(k_a^{\text{dep}}, k_b^{\text{dep}}),\, 0\big)$, and $H_d = \sum_{\mathcal{P}_d} h_{ab}$.

3. $\mathcal{S}_d$ is the set of the surrogate pairs (i, j, r); S_d is their exposure, the same sum with j shifted: $S_d = \sum_{\mathcal{S}_d} \max\big(\min(k_i^{\text{fin}}, k_j^{\text{fin}} - r) - \max(k_i^{\text{dep}}, k_j^{\text{dep}} - r),\, 0\big)$.

4. $M_d(\tau)$ is the number of contact intervals of the surrogate pairs lasting at least τ, and $\rho_{0,d}(\tau) = M_d(\tau)/S_d$ the chance intervals per second of exposure.

Over the archive, with $N(\tau)$ the number of real intervals of the pairs of every $\mathcal{P}_d$ lasting at least τ:

$$C(\tau) = \sum_d \rho_{0,d}(\tau)\, H_d, \qquad \widehat{\text{FDR}}(\tau) = \frac{C(\tau)}{N(\tau)}, \qquad \tau^* = \min\big\{\tau :\ \widehat{\text{FDR}}(\tau) \le q\big\}. \tag{14}$$

Dates without surrogates add to N only, and τ runs over the grid $\{0, 5, \ldots, 14\,400\}$ s. The value $q = 0.1$ is chosen in Section 8.4.

> **In words.** $C(\tau)$ is the number of intervals lasting at least τ that chance alone would give the real pairs. Dividing it by the number actually observed gives the share of chance intervals among them; τ^* is the shortest duration at which that share falls to q, one in ten.

This is the estimate of the false discovery rate of SAM [26], the number of discoveries expected by chance over the number observed, with time shifts in place of the permutations.

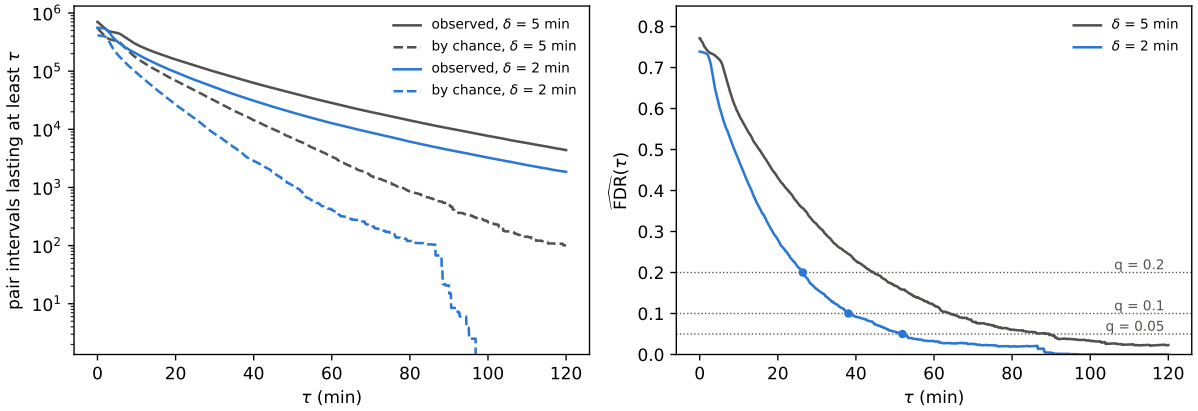


Figure 28: Left: real pair intervals lasting at least τ (solid) and those chance alone gives (dashed), with $\delta = 5$ min (grey) and $\delta = 2$ min (blue), on the whole archive. Right: their ratio, the false discovery rate of Eq. (14); dots mark τ^* for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

Most contacts are chance (Figure 28): 74% of all pair intervals with $\delta = 2$ min (77% with 5 min), 47% of those lasting 10 min, 16% of those lasting 30 min. Short contacts therefore say little; only long ones are mostly real. This is why τ^* is tens of minutes: it is the duration at which the real contacts outnumber chance by the factor $1/q$. Resampling the dates 200 times moves τ^* by a few minutes (Table 10), so the curve, and not the noise of the surrogates, sets it.

7.3 How much the shift matters

The false discovery rate was recomputed with only the surrogate pairs of each date whose shift falls in one class, each class with its own exposure S_d and intervals. The classes are compared on the 2 737 dates (of 6 209) that have surrogates in each of them (Table 6).

Surrogates	pairs	$\widehat{\mathrm{FDR}}$(10 min)	$\widehat{\mathrm{FDR}}$(30 min)	τ^* (min)	5–95% over dates
all shifts	241 824	0.48	0.16	38.2	37.0–41.5
$\lvert r\rvert < 30$ min	69 444	0.73	0.31	53.3	49.7–72.0
$\lvert r\rvert \geq 60$ min	119 687	0.26	0.05	23.0	21.1–24.4

Table 6: The false discovery rate and τ^* ($q = 0.1$, $\delta = 2$ min) with the surrogates of one class of shifts, on the 2 737 dates with surrogates in each class.

Short shifts give 2.8 times the chance of long ones at 10 min and 5.7 times at 30 min, and τ^* goes from 23 min with shifts of an hour or more to 53 min with shifts under 30 min; with all the shifts it is 38 min, as on the whole archive. Short shifts keep the weather of the hour, but they also bring within δ the pilots who follow each other a few minutes apart, and the delay profile shows that these are many: real correspondences are still 2 to 3 times chance between 195 and 500 s (Section 8.1). The two effects cannot be separated here. The length of the shift decides what counts as chance and moves τ^* by more than a factor two; we keep the shifts that follow the takeoff delays of the day.

8 The parameters on the data

The method has three kinds of parameters (Table 7). The *tolerances* say when two positions match: δ, ε_h, ε_z, $\theta_{\max}$, g. The *duration* τ^* says how long a match must last to be a contact; it follows from the share q of chance contacts we accept (Section 7.2). The other *settings* make the computation exact or fast and are fixed by physical bounds. Each tolerance is compared with chance through the null model of Section 7.1: the profiles of this section use a sample of 300 random dates, the choices of δ and q the whole archive.

Parameter	Value	How it was set
δ, largest delay	2 min	measured on the delay profile, Section 8.1
q, accepted false discovery rate	0.1	trade-off, Section 8.4
τ^*, shortest contact	38.1 min	computed from q and the null model, Section 7.2
ε_h, ε_z	500 m, 300 m	definitions: profiles without an edge, Section 8.2
g, longest gap joined	120 s	definition: gaps of real and chance contacts alike, Section 8.2
$\theta_{\max}$	45°	no effect on the result, Section 8.3
Δ, $h_{\max}$, R_{dep}, $v_{\max}$, R_s, $r_{\min}$	see Section 8.5	physical bounds and exactness

Table 7: The parameters and how each was set.

8.1 The maximum delay δ

Section 4.4 bounds δ by the visual range, at about 5 min. Here the data say how long the delay between two pilots who fly together actually is.

The delay profile. For every delay l, the correspondences $(t, t+l)$ of the real pairs and of the surrogate pairs were counted, per second of exposure, with a window of 15 min on 300 random dates (58 910 h of real exposure):

$$R(l) = \frac{\text{real correspondences at delay } l \text{ per second of exposure}}{\text{chance correspondences at delay } l \text{ per second of exposure}}. \tag{15}$$

Chance is flat in the delay, as it should: a shifted copy has no preferred delay. The real pairs peak at a few seconds, where they have 5.6 times the correspondences of chance, and decay: R falls below 4 at 105 s, below 3 at 195 s and below 2 at 500 s, and is still 1.5 at 15 min (Figure 29). The excess has two parts. The fast one, within 2–3 min, is pilots flying together. The slow one, which has not ended at 15 min, is pilots who use the same thermals at about the same time without following each other, a synchrony that the shifts of the surrogates remove. A rule such as "the delay that holds 95% of the excess" would pick 13 min, because it counts the slow tail; that is why we read the ratio instead.

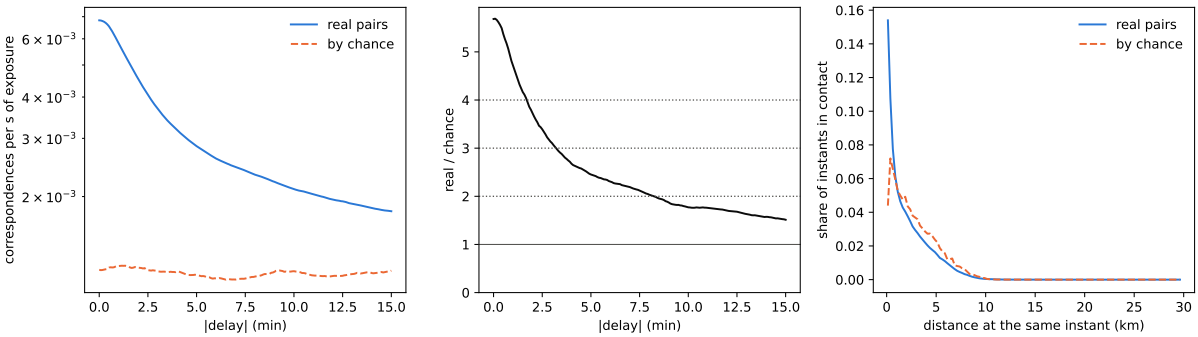


Figure 29: The delay profile on 300 dates with a 15 min window. Left: correspondences per second of exposure at each delay, real and by chance. Middle: their ratio $R(l)$ of Eq. (15), with the levels 4, 3 and 2. Right: the distance between the two pilots at the same instant when one is on the path of the other.

The delay inside the contacts. The profile counts correspondences; this view looks at the contacts themselves, through the delay ℓ_{med} of each pair interval (Section 4.5). Its absolute value was computed for every pair interval of the run with $\delta = 5$ min, whose window is wide enough to show where the delays fall: 965 770 intervals, of which 169 517 have all their instants in search and no delay.

Duration of the interval	intervals	median $\lvert\ell_{\text{med}}\rvert$	within 2 min	at the edge
under 10 min	436 575	210 s	32%	29%
10–30 min	246 587	160 s	39%	12%
30–90 min	102 357	80 s	64%	2.1%
90 min or more	10 734	25 s	90%	6 intervals

Table 8: The delay of the pair intervals by their duration ($\delta = 5$ min, whole archive). At the edge: $\lvert\ell_{\text{med}}\rvert \geq 285$ s, the last 15 s of the window.

The longer the contact, the shorter the delay (Table 8, Figure 30). A pair that only shares route and weather matches at any delay of the window, and its closest correspondence is often at the largest ones: short intervals pile up at the edge, 29% of those under 10 min. The contacts of at least 90 min, all above τ^* of this run (89.5 min), have a median delay of 25 s; 90% of them lie within 2 min and only 6 of 10 734 at the edge. Long contacts are pilots a few tens of seconds apart on the same path.

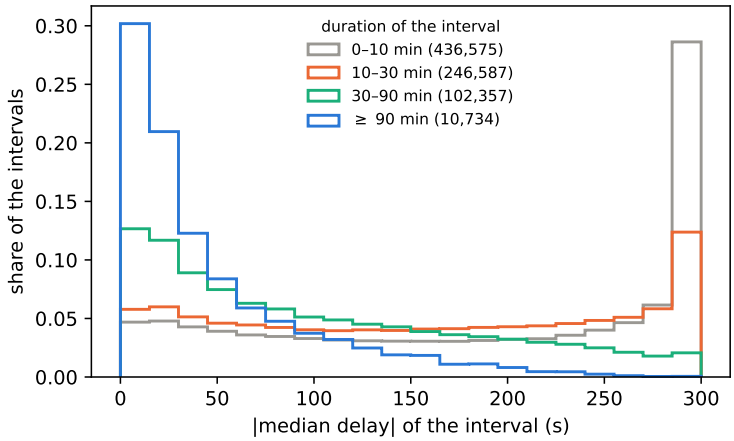


Figure 30: Distribution of the $|\ell_{\text{med}}|$ of the pair intervals, by their duration ($\delta = 5$ min, whole archive).

Choice. We take $\delta = 2$ min: within it the real correspondences are at least 3.7 times chance, and it contains nine in ten of the long contacts. Shortening the delay also shortens τ^*, since fewer chance matches enter: from 89.5 to 51.9 min at $q = 0.05$ (Table 10).

Distance at the same instant. The delay also bounds how far apart two pilots in contact can be at the same instant: at most $\varepsilon_h + v_{\max}\delta \approx 0.5 + 27.8 \times 120/1000 \approx 3.8$ km. A separation of 6 km, the largest we want to call a group, is never reached by one pair; a group, a chain of pairs, can be wider. Even with the 15 min window, 58% of the instants in contact have the two pilots within 2 km and 94% within 6 km (Figure 29, right).

The same distance was also counted without requiring a correspondence: at every grid instant when both fly after departure, for all the real pairs and for the surrogates, on the dates of Figure 29 that have surrogates (210 of 300, 58 901 h of real exposure). Real pairs are much closer than chance at short distance and have no edge (Figure 31): the ratio $g(d)$ is 10.4 within 100 m, below 2 from 500 m, below 1.5 from 900 m, below 1.1 from 2.7 km and below 1 from 5.7 km. Real pairs spend 6.6% of their time together within 1 km of each other, against 2.8% by chance. Beyond 6 km g falls slightly below 1, because both distributions sum to one. The excess reaches several kilometres, beyond the visual range of 1.5 km of Section 4.4. Since the surrogates also change the hour, this excess may hold the weather of the hour as well as sight, so it bounds neither the visual range nor the range of following.

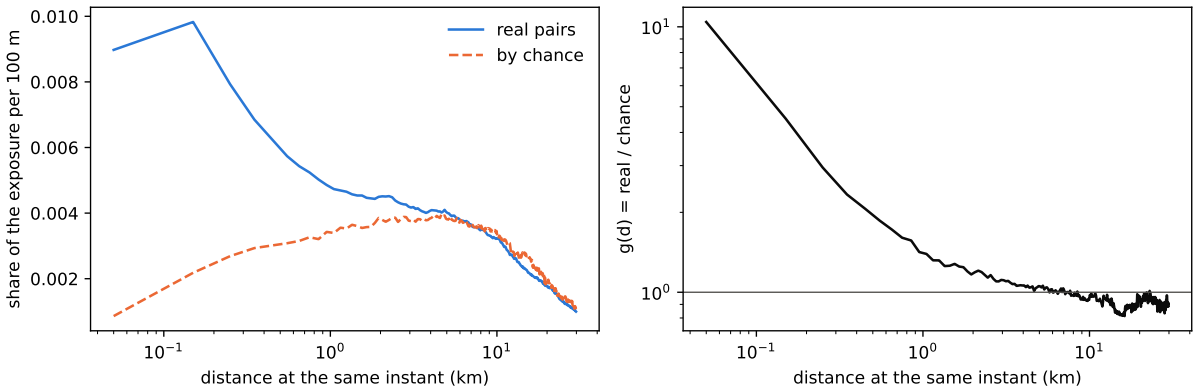


Figure 31: Distance between the two pilots at the same instant, over all the time both fly, on 210 dates. Left: share of the exposure in each 100 m bin, real pairs and chance. Right: their ratio $g(d)$.

8.2 The tolerances against chance

In analogy with the delay profile, with $\delta = 2$ min and the null model of the chosen run, at every instant t of pilot i on the 210 dates of Figure 31:

- the smallest horizontal distance between i at t and j at $t + l$, $|l| \leq \delta$, over the delays with $|z_i(t) - z_j(t + l)| < \varepsilon_z$ and headings that pass (left of Figure 32);
- the smallest height difference over the delays with horizontal distance $< \varepsilon_h$ and headings that pass (middle);
- the gaps between consecutive instants of contact of a pair (right).

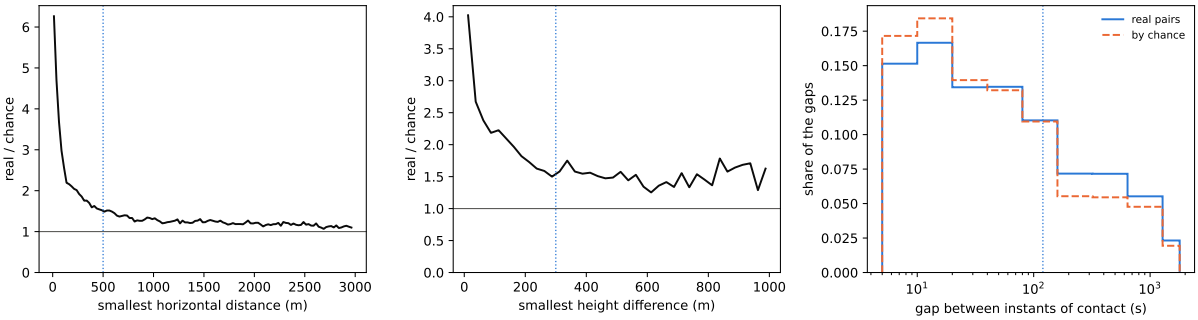


Figure 32: The tolerance profiles on 210 dates. Left and middle: real pairs over chance, per second of exposure, by the smallest horizontal distance and the smallest height difference to the path of the other pilot within δ; dotted, the values in use. Right: the lengths of the gaps between instants of contact, as a share of all gaps (the 8% of real and 9% of chance gaps longer than 30 min are not drawn); dotted, g.

None of the three has an edge (Figure 32). The horizontal ratio is 6.3 within 25 m, below 2 from 250 m, below 1.5 from 500 m, and still 1.13 between 2.5 and 3 km. The vertical ratio is 4.0 within 25 m and below 2 from 150 m; from 275 m on it stays near 1.5 (1.51 on average up to 1 km), the excess that the horizontal condition alone gives. Both values in use sit where real pairs are about 1.5 times chance.

Real pairs have 1.46 times the gaps of chance per hour, because they have more contact, but the lengths of the gaps are the same: the shares of each class differ by at most 2 points. The gaps come from how pilots fly in and out of each other's path, the same in real and in chance contacts, so the data do not single out g.

8.3 One parameter at a time

A plateau is a range of values of the tolerances around which the result does not change: a value inside it makes the result insensitive to the exact threshold. If there is none, the thresholds decide the result, and this must be said. The result is measured by f_G, the fraction of the flight time after the departure spent in a group:

$$f_G = \frac{\sum_i \left|\{t \in T_i^{\text{dep}} : \ \gamma_i(t) \notin \{\mathsf{L}, \mathsf{S}\}\}\right|}{\sum_i |T_i^{\text{dep}}|}. \tag{16}$$

We changed one parameter at a time on 300 random dates, recalibrating τ^* each time, with $\delta = 5$ min (Table 9). $\theta_{\max}$ changes nothing: contacts in glide are few. The others move f_G by up to a factor three, mostly through τ^*: there is no plateau, in agreement with the profiles, and ε_h, ε_z and g are definitions. On 300 dates τ^* is noisy (126 min with the starting values against

89.5 on the whole archive, and six settings stop at the same surrogate interval); a conclusive check needs the whole archive for each setting, about one hour each. It was not repeated with $\delta = 2$ min.

Setting ($\delta = 5$ min unless stated)	τ^* (min)	f_G	episodes	flights in a group
starting values	126.1	0.034	98	210
$\varepsilon_h = 250$ m	75.1	0.058	253	537
$\varepsilon_h = 1000$ m	126.8	0.065	162	372
$\varepsilon_z = 150$ m	126.1	0.020	61	125
$\varepsilon_z = 600$ m	126.1	0.044	122	268
$\delta = 2$ min	42.6	0.106	600	1 170
$\delta = 10$ min	128.4	0.062	153	365
$\theta_{\max} = 30°$	126.1	0.034	97	208
$\theta_{\max} = 90°$	126.1	0.034	98	210
$g = 60$ s	75.8	0.087	332	732
$g = 300$ s	126.1	0.055	141	318

Table 9: One parameter at a time on 300 random dates (7 051 flights), $q = 0.05$.

8.4 The false discovery rate q

q is the share of chance contacts we accept among those we keep. A smaller q keeps only very long contacts and few groups, all of them real; a larger q sees more groups, with more chance among them. Table 10 and Figure 33 give the trade-off.

δ	q	τ^* (min)	5–95% over dates	para in a group	hang in a group	episodes
5 min	0.05	89.5	79.5–91.5	8.9%	2.1%	6 090
2 min	0.05	51.9	49.1–53.3	13.7%	3.8%	10 553
2 min	0.1	38.1	37.0–41.1	21.7%	6.9%	17 169
2 min	0.2	26.4	25.2–27.2	31.7%	13.9%	26 504

Table 10: τ^* and the groups it gives. Flights with a logger of at most 10 s and a position after departure: 154 598 paragliders and 6 039 hang gliders.

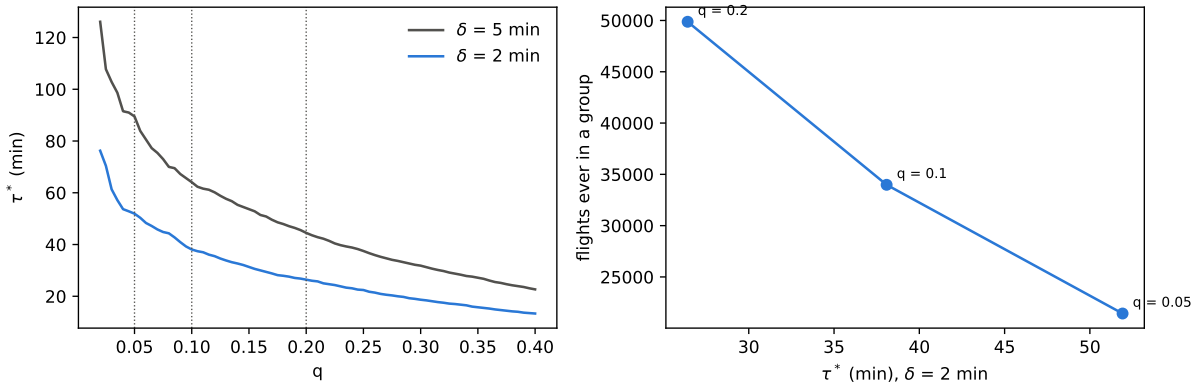


Figure 33: Left: τ^* as a function of q, for the two delays. Right: flights ever in a group against τ^*, for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

We take $q = 0.1$: at most one contact in ten is chance, and $\tau^* = 38.1$ min. Going to 0.2 would lower τ^* to 26 min but would let chance make one group in five.

8.5 Fixed settings

- $\Delta = 5$ s, the common grid: a pilot is at most 69 m from a grid position at 100 km/h, against $\varepsilon_h = 500$ m (Section 6.2).
- $h_{\max} = 10$ s: positions are interpolated only between fixes at most 10 s apart; the flights of coarser loggers are left out of the statistics.
- $R_{\text{dep}} = 3$ km: contacts around the takeoff, before the first exit from this disc, are not counted (Section 4.1).
- $v_{\max} = 100$ km/h, the speed of the candidate filter: pieces of track end at every faster step, so the filter never misses a correspondence (Lemma 6.1, checked against every pair of instants on synthetic days). The profiles of Section 8.2 widen the filter to 3 km, the largest distance they count.
- $R_s = 10$ km, $r_{\min} = 2\delta + \Delta$, at most 200 surrogate pairs per date: the null model of Section 7.1.

9 The groups on the archive

With the chosen values, $\delta = 2$ min and $q = 0.1$, 21.7% of the paraglider flights and 6.9% of the hang-glider flights fly in a group at some point, in 17 169 episodes (Table 10). This section describes those groups. A configuration is a group with fixed members; it ends when a member joins or leaves. Its duration runs from its start to its end, and its participants are its members at once. An episode is a chain of configurations; its duration runs from the start of its first configuration to the end of its last, and its participants are the distinct flights it holds. The participants are counted in two ways: the share of the episodes (or configurations) of each size, and the share of their total duration.

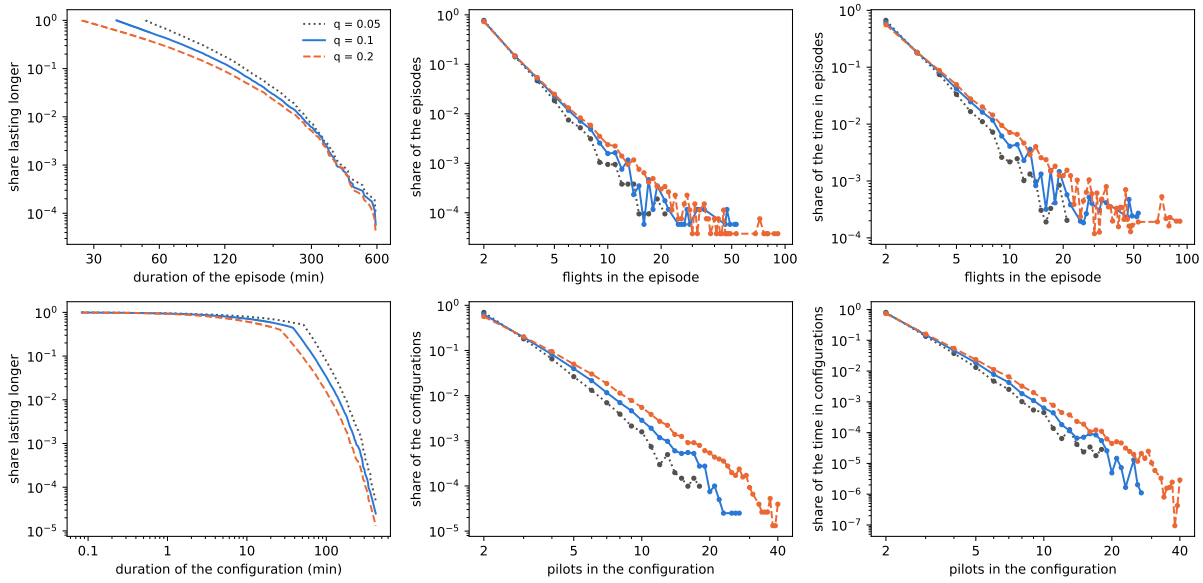


Figure 34: Episodes (top) and configurations (bottom), $\delta = 2$ min, whole archive, for three values of q. Left: share lasting longer than a duration. Middle: share of each size. Right: share of the total duration in each size.

	q	number	median duration	2 participants items	2 participants time	5 or more items	5 or more time	largest
episodes	0.05	10 553	74.6 min	77%	67%	3.9%	8.1%	21
	0.1	17 169	58.2 min	74%	61%	5.7%	12.6%	53
	0.2	26 504	44.5 min	73%	56%	7.0%	17.2%	91
configurations	0.05	20 275	52.7 min	70%	81%	5.6%	2.3%	18
	0.1	40 051	30.4 min	63%	77%	9.4%	3.5%	27
	0.2	75 953	17.1 min	56%	74%	14.1%	5.0%	40

Table 11: Episodes and configurations for three values of q ($\delta = 2$ min). Items: share of the episodes or configurations of that size; time: share of their total duration.

With the chosen values the median episode lasts 58.2 min and three in four are pairs (Table 11). Episodes last at least τ^*, since a contact must. Weighted by duration the pairs weigh less, 61% of the episode time, because larger episodes last longer: those of five flights or more are 5.7% of the episodes and 12.6% of their time.

Configurations have no minimum duration: every arrival or departure of a member starts a new one, so 10% last less than 2.1 min, and the median is 30.4 min. Weighted by duration the pairs weigh more, 77% of the time against 63% of the configurations, because large configurations change quickly.

A larger q, hence a shorter τ^*, gives more and shorter groups, and larger ones: the largest episode holds 21 flights with $q = 0.05$, 53 with 0.1 and 91 with 0.2. The shares of the small sizes change little; the tail moves.

10 Outputs

Table	Columns (type) and content
T1 `pair_intervals`	`date`, `a`, `b` (int32), `t_on`, `t_off` (int32, s), `duration` (int32), `ell_med` (int16, s, signed: positive if `b` follows `a`), `d_dec` (float32, m). Who was with whom, and when.
T2 `configurations`	`date`, `conf_id`, `episode_id` (int32), `t_start`, `t_end` (int32), `n` (int16), `members` (list of int32).
T3 `membership`	`flight` (int32), `t_start`, `t_end` (int32), `state` (int8: L, S, G), `conf_id`, `episode_id` (int32, null if L or S), `n` (int16, 1 if S). The main table: each flight cut into stretches of constant membership.
T4 `episodes`	`episode_id`, `date`, `t_start`, `t_end`, `n_flights`, `n_max`, `n_config`, `block`.
T5 `flight_summary`	`flight`, `phi_G` (fraction of T_i^{dep} in a group), `kappa` (distinct companions), `n_entries`, `n_episodes`, `block`, `local` (boolean).

Table 12: Final tables, in Parquet. The members of a configuration are not repeated in T3: they are read from T2 through `conf_id`.

The state of flight i at instant t is obtained with an as-of join on (`flight`, t) in T3, sorted by `t_start` (Figure 35). T3 has about $N_{\text{flights}}(2 + 2\bar{c})$ rows, with $\bar{c}$ the mean number of configurations per flight: of the order of 10^5–10^6 rows, tens of MB. The other tables are smaller.

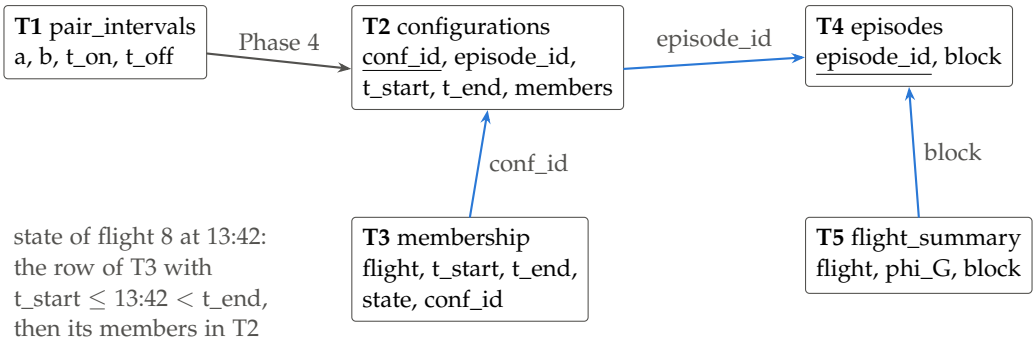


Figure 35: How the tables connect. Arrows point from a column to the table where it is a key. The example reads the state of a flight at one instant with an as-of join on T3.

11 Use for the comparison between group and solo flight

Labels on the phases. For a run σ of one phase (climb, search or glide) of flight i, with interval J_σ, define the fraction in a group and the mean size:

$$f_\sigma = \frac{\left|\{t \in J_\sigma \cap T_i^{\text{dep}} : \gamma_i(t) \notin \{\mathsf{L},\mathsf{S}\}\}\right|}{|J_\sigma \cap T_i^{\text{dep}}|}, \qquad \bar{n}_\sigma = \text{mean of } n_i(t) \text{ over } J_\sigma \cap T_i^{\text{dep}}, \tag{17}$$

with $n_i(t)$ the number of members of the configuration of i at t, 1 if solo. The run is *in a group* if $f_\sigma \geq 1/2$ and *solo* if $f_\sigma = 0$; the intermediate cases are excluded, or f_σ is used as a continuous covariate (Figure 36). Runs in the local phase are excluded.

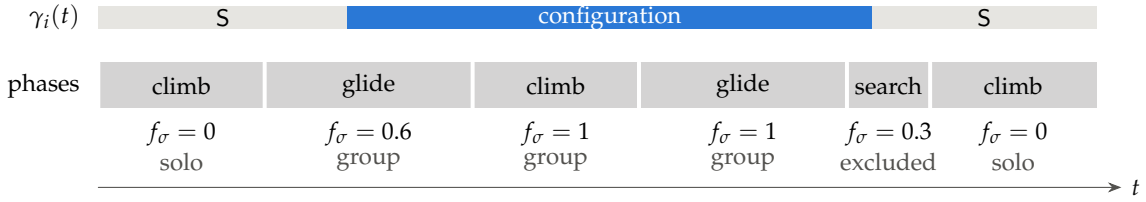


Figure 36: Labels on the phases of one flight. Each run of one phase gets the fraction f_σ of its duration spent in a configuration: runs with $f_\sigma \geq 1/2$ count as group, those with $f_\sigma = 0$ as solo, the search with $f_\sigma = 0.3$ is left out.

Stratification. Group flights concentrate on the good days and the busy sites: a global comparison would mix the effect of the group with that of the weather. The comparison runs within strata s (date and area):

$$\hat{\Delta} = \sum_s w_s \left(\hat{\theta}_{G,s} - \hat{\theta}_{S,s}\right), \tag{18}$$

where $\hat{\theta}$ is the statistic of interest and the weights w_s are proportional, for example, to the number of group runs of the stratum.

Uncertainties. Runs of pilots of the same episode are not independent, and neither are those of the same flight. The bootstrap resamples the dependency blocks (Section 5.5) within the strata.

Confounders. Competition days; the level of the pilot (the same pilots often fly together); the glider class (EN A–D); the time of day.

Questions to ask. Distributions of the times spent in thermals and of the lengths of the glides, in a group and alone; mean squared displacement and effective exponent; differences between followers and leaders, by the sign of ℓ_{med}. In birds, followers leave the thermals earlier and lower [8], and individuals with social information glide faster [28]: these are hypotheses to test on pilots before assuming them.

12 Choices to declare and limits

The starting values and those chosen on the data are in Table 3. The thesis should state these choices, each argued in the section given:

- ε_z encodes the notion of "together": shared thermals if tight, visual contact if wide (Section 4.4);
- δ is bounded by the visual range through v_{xc} (Section 4.4) and measured on the data at 2 min (Section 8.1);
- $q = 0.1$: at most one contact in ten is chance (Section 8.4);
- ε_h, ε_z and g show no plateau on the data, so their values are definitions (Sections 8.2 and 8.3);
- headings are compared only when both pilots glide; climb, search and unclassified fixes turn the condition off (Section 4.3); in the run reported here the phases are Vilpellet's, on 1 Hz flights only, so the condition acts only between two such flights (Section 3);
- contacts before the departure are ignored: a group formed on the takeoff is recognised only after the exit from R_{dep} (Section 4.1);
- grid positions are interpolated across at most $h_{\max}$, so tracks logged more coarsely enter with gaps (Section 6.2);
- the signed median delay leaves out the instants of search, where the delay is not determined (Section 4.5);
- chance is defined by the null model, whose shifts of about an hour also remove the synchrony of pilots with the weather of the hour: what it calls real includes pilots who share the same thermals at the same time without following each other, the "same air" notion of Remark 4.3, which is not proof of interaction. τ^* is 23 min with shifts over an hour and 53 min with shifts under 30 min (Section 7.3);
- the errors of the phase labelling reach the heading condition: a climb labelled as glide compares headings that turn, and joining the short gaps mitigates the problem.

The measurements have three further limits:

- the profiles of Sections 8.1 and 8.2 are on one sample of 300 dates;
- the plateau check of Section 8.3 is on 300 dates and with the starting delay of 5 min;
- the data are the cleaned tracks of the old repository (soaring, tag `legacy-final`), not the pre-processing of xc-dynamics that Section 3 describes.

12.1 Other checks

- Visual inspection: about ten episodes on a map, with the entries and exits checked by hand.
- Sensitivity to the step: on a subset of pairs, the computation with $\Delta = 1$ s must give the same intervals.
- Competition days: if the dataset contains competition tracks, on those days everybody flies the same course and is in a group by construction; they are flagged and treated apart.
- Grid: the share of grid instants without a position, per flight and by logging interval.

A Why $m_{ij}(t) = 1$ does not give $m_{ji}(t) = 1$

$m_{ij}(t)$ of Definition 4.2 asks whether i, at instant t, is at a place that j visits within δ; $m_{ji}(t)$ asks whether j, at the same instant, is at a place that i visits within δ. These are two questions about two different pairs of instants. Both are read from one set of correspondences $\mathcal{M}_{ij}$, one on each axis, and they agree only if that set is symmetric about the diagonal $t' = t$. What does hold:

- if $m_{ij}(t) = 1$ through the pair (t, t'), then $m_{ji}(t') = 1$: the same pair, read on the other axis, $l = t' - t$ later;
- every instant of one indicator lies within δ of an instant of the other;
- the contact $a_{ij} = m_{ij} \vee m_{ji}$ is symmetric: $a_{ij} = a_{ji}$.

This appendix keeps only the horizontal condition of $\mathcal{M}_{ij}$: the vertical and heading conditions only remove pairs. m_{ij} fixes the instant of i and lets that of j vary within $\pm\delta$; m_{ji} fixes the instant of j and lets that of i vary. Nothing makes the two answers equal.

A.1 A worked example

Take $\delta = 120$ s and $\varepsilon_h = 500$ m, and ignore heights and headings. Pilot i circles in a thermal at the point P from 0 to 600 s; the radius of the circle is negligible against ε_h. Pilot j flies east in a straight line at 10 m/s and is in P at 500 s.

t (s)	j relative to P	m_{ij}	m_{ji}	a_{ij}	what happens
300	2.0 km west	0	0	0	j needs more than 120 s to come within 500 m of P
380	1.2 km west	1	0	1	i is where j will be 120 s later; j is not yet where i has been
500	in P	1	1	1	both are in P
560	0.6 km east	1	0	1	i is where j was 60 s earlier; j has left

Table 13: The four instants of the example. Over the whole flight $m_{ij} = 1$ from 331 to 600 s, $m_{ji} = 1$ from 451 to 549 s, and $a_{ij} = 1$ from 331 to 600 s (strict inequality: exactly 500 m does not count).

A.2 The same example in the plane (t, t')

Plot every correspondence (t, t') with the instant at which i passes across and the one at which j passes up (Figure 38a). Then $m_{ij}(t) = 1$ exactly when the *vertical* line at t meets $\mathcal{M}_{ij}$, and

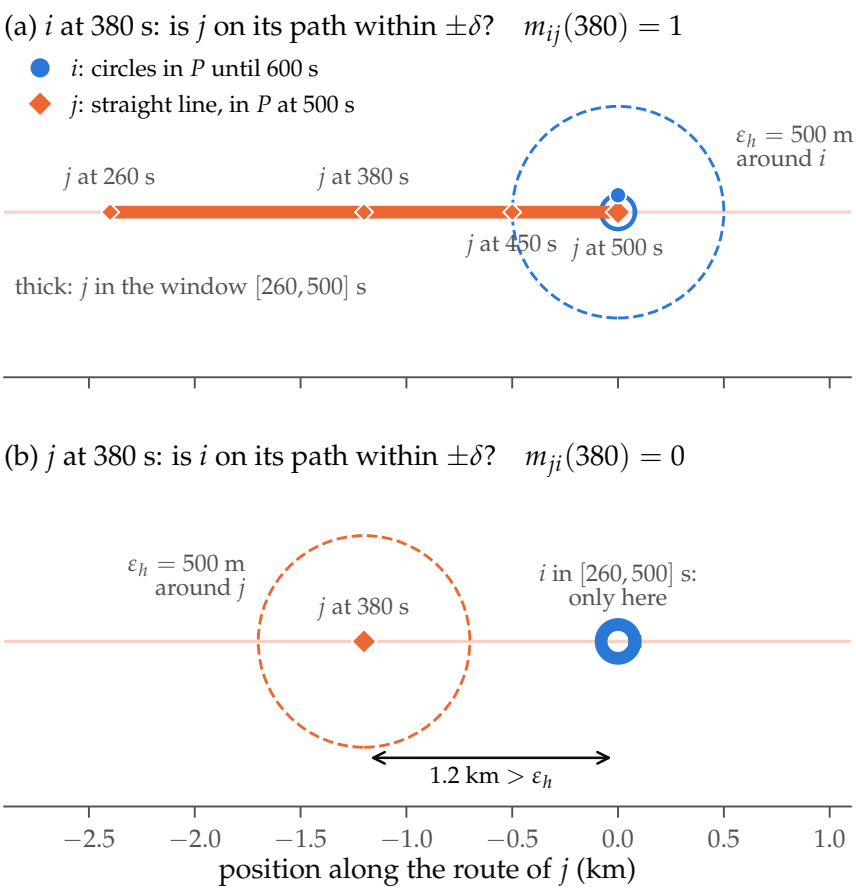


Figure 37: The example at 380 s. (a) i is in P and j is 1.2 km west of it. In the window $[260, 500]$ s of $\pm\delta$ (thick), j comes within ε_h of i at 451 s and is in P at 500 s: the pair $(380, 500)$ is a correspondence, so $m_{ij}(380) = 1$. (b) At the same instant j is 1.2 km from P, and during its window i has been only in P: no position of i lies within ε_h of j, so $m_{ji}(380) = 0$. The ring only marks P.

$m_{ji}(t) = 1$ exactly when the *horizontal* line at height t meets it. The two indicators are the shadows of $\mathcal{M}_{ij}$ on the two axes.

At $t = 380$ the vertical line meets $\mathcal{M}_{ij}$ (at the pair $(380, 500)$ and the ones just below it), and the horizontal line does not: $\mathcal{M}_{ij}$ has no point with $t' < 451$. The set is not symmetric about the diagonal. The mirror of the pair $(380, 500)$ would be $(500, 380)$, which needs i at 500 s within 500 m of j at 380 s, and that is 1.2 km away.

A.3 How far apart they can be

Claim. If $m_{ij}(t) = 1$ there is an instant t' with $|t' - t| \le \delta$ and $m_{ji}(t') = 1$; and conversely.

Proof. Take the pair $(t, t') \in \mathcal{M}_{ij}$ that gives $m_{ij}(t) = 1$. Then $|t' - t| \le \delta$ by the definition of $\mathcal{M}_{ij}$, and $m_{ji}(t') = 1$ by the definition of m_{ji}. The converse is the same argument read the other way. □

So the two indicators can differ only by a shift of at most δ. The bound is reached in the example: the first instant of m_{ij} (331 s) is exactly δ before the first instant of m_{ji} (451 s). The union a_{ij} inherits it: at each end of a contact it extends the shorter indicator by up to δ.

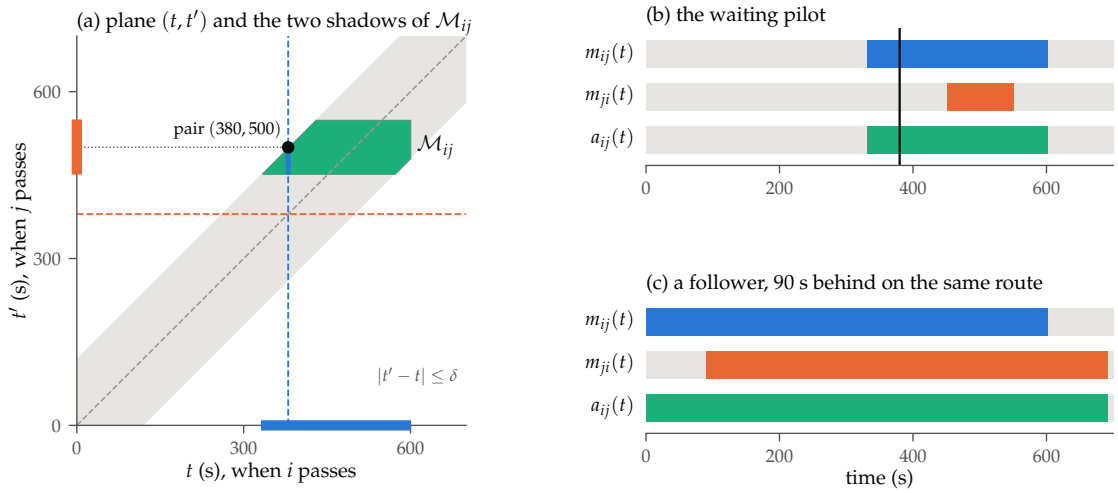


Figure 38: (a) The correspondences $\mathcal{M}_{ij}$ (green) in the band $|t' - t| \le \delta$, and its shadows m_{ij} (blue bar on the t axis) and m_{ji} (orange bar on the t' axis). The vertical line at 380 s meets $\mathcal{M}_{ij}$ (thick segment), the horizontal one does not. (b) The indicators over time, with the line at 380 s. (c) A pilot j on the same route 90 s behind i: m_{ji} is m_{ij} delayed by exactly 90 s.

definition	symmetric	in the example	remark
$m_{ij} \vee m_{ji}$ (used)	yes	331–600 s	counts the approach and the departure within δ
$m_{ij} \wedge m_{ji}$	yes	451–549 s	shorter: each on the path of the other at once
m_{ij} alone	no	331–600 s, or 451–549 s for (j, i)	depends on which pilot is called i

Table 14: Three ways to define the contact of a pair.

A.4 Why the contact is the union

The union makes the contact independent of the order of the two pilots and keeps the approach and the departure, where one of them is already on the path of the other. The price is that a contact starts up to δ before the pilots are close at the same time. The mutual version $m_{ij} \wedge m_{ji}$ would give shorter contacts: the false discovery rate and τ^* would have to be computed again with the null model, which is not done here.

In the code. `contacts.scan` returns the instants of $m_{ij} = 1$ and, separately, those of $m_{ji} = 1$; `contacts.pair_intervals` takes their union. Scenarios: `models.WAIT`, `models.FOLLOW`; figures: `plots.asymmetry`; numbers checked in `test_note.py`.

References

[1] Z. Ákos, M. Nagy, T. Vicsek. Comparing bird and human soaring strategies. *Proceedings of the National Academy of Sciences* 105(11):4139–4143, 2008.

[2] C. Amornbunchornvej, I. Brugere, A. Strandburg-Peshkin, D. R. Farine, M. C. Crofoot, T. Y. Berger-Wolf. Coordination event detection and initiator identification in time series data. *ACM Transactions on Knowledge Discovery from Data* 12(5), 2018. doi:10.1145/3201406.

[3] C. Amornbunchornvej, T. Y. Berger-Wolf. Framework for inferring leadership dynamics of complex movement from time series. *Proceedings of the SIAM International Conference on Data Mining*, 549–557, 2018.

[4] C. Amornbunchornvej. mFLICA: an R package for inferring leadership of coordination from time series. *SoftwareX*, 2021. doi:10.1016/j.softx.2021.100781. Preprint arXiv:2004.06092.

[5] M. Andersson, J. Gudmundsson, P. Laube, T. Wolle. Reporting leadership patterns among trajectories. *Proceedings of the ACM Symposium on Applied Computing*, 3–7, 2007.

[6] K. Buchin, M. Buchin, M. van Kreveld, B. Speckmann, F. Staals. Trajectory grouping structure. *Journal of Computational Geometry* 6(1), 2015. doi:10.20382/jocg.v6i1a3. Preliminary version: WADS 2013; arXiv:1303.6127.

[7] L. Chen, M. T. Özsu, V. Oria. Robust and fast similarity search for moving object trajectories. *Proceedings of ACM SIGMOD*, 491–502, 2005.

[8] A. Flack, M. Nagy, W. Fiedler, I. D. Couzin, M. Wikelski. From local collective behavior to global migratory patterns in white storks. *Science* 360(6391):911–914, 2018.

[9] D. Greene, D. Doyle, P. Cunningham. Tracking the evolution of communities in dynamic social networks. *Proceedings of the International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 176–183, 2010.

[10] J. Gudmundsson, M. van Kreveld. Computing longest duration flocks in trajectory data. *Proceedings of ACM GIS*, 2006.

[11] H. Jeung, M. L. Yiu, X. Zhou, C. S. Jensen, H. T. Shen. Discovery of convoys in trajectory databases. *Proceedings of the VLDB Endowment* 1(1), 2008.

[12] R. Joo, M.-P. Etienne, N. Bez, S. Mahévas. Metrics for describing dyadic movement: a review. *Movement Ecology* 6:26, 2018.

[13] M. Konzack, T. McKetterick, T. Ophelders, M. Buchin, L. Giuggioli, J. Long, T. Nelson, M. A. Westenberg, K. Buchin. Visual analytics of delays and interaction in movement data. *International Journal of Geographical Information Science* 31(2):320–345, 2017.

[14] Z. Li, B. Ding, J. Han, R. Kays. Swarm: mining relaxed temporal moving object clusters. *Proceedings of the VLDB Endowment* 3(1), 2010.

[15] Z. Li, F. Wu, M. C. Crofoot. Mining following relationships in movement data. *Proceedings of the 13th IEEE International Conference on Data Mining (ICDM)*, 458–467, 2013. doi:10.1109/ICDM.2013.98.

[16] J. A. Long, T. A. Nelson, S. L. Webb, K. L. Gee. A critical examination of indices of dynamic interaction for wildlife telemetry studies. *Journal of Animal Ecology* 83(5):1216–1233, 2014.

[17] J. A. Long, S. L. Webb, S. M. Harju, K. L. Gee. Analyzing contacts and behavior from high frequency tracking data using the wildlifeDI R package. *Geographical Analysis* 54:648–663, 2022.

[18] M. Nagy, Z. Ákos, D. Biro, T. Vicsek. Hierarchical group dynamics in pigeon flocks. *Nature* 464:890–893, 2010.

[19] M. Nagy, I. D. Couzin, W. Fiedler, M. Wikelski, A. Flack. Synchronization, coordination and collective sensing during thermalling flight of freely migrating white storks. *Philosophical Transactions of the Royal Society B* 373:20170011, 2018.

[20] W. L. Ruzzo, M. Tompa. A linear time algorithm for finding all maximal scoring subsequences. *Proceedings of ISMB*, 1999.

[21] H. Sakoe, S. Chiba. Dynamic programming algorithm optimization for spoken word recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 26(1):43–49, 1978.

[22] O. Spiegel, S. T. Leu, A. Sih, C. M. Bull. Socially interacting or indifferent neighbours? Randomization of movement paths to tease apart social preference and spatial constraints. *Methods in Ecology and Evolution* 7(8):971–979, 2016.

[23] H. Su, S. Liu, B. Zheng, X. Zhou, K. Zheng. A survey of trajectory distance measures and performance evaluation. *The VLDB Journal* 29:3–32, 2020.

[24] R. E. Tarjan. Efficiency of a good but not linear set union algorithm. *Journal of the ACM* 22(2):215–225, 1975.

[25] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272, 2020.

[26] V. G. Tusher, R. Tibshirani, G. Chu. Significance analysis of microarrays applied to the ionizing radiation response. *Proceedings of the National Academy of Sciences* 98(9):5116–5121, 2001.

[27] M. Vlachos, G. Kollios, D. Gunopulos. Discovering similar multidimensional trajectories. *Proceedings of the 18th International Conference on Data Engineering (ICDE)*, 2002.

[28] H. J. Williams, A. J. King, O. Duriez, L. Börger, E. L. C. Shepard. Social eavesdropping allows for a more risky gliding strategy by thermal-soaring birds. *Journal of the Royal Society Interface* 15:20180578, 2018.